ELARA: Reliability-Aware Multimodal Anomaly Fusion Under Degraded Evidence

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Abstract*—Multimodal anomaly systems often fail not because domain experts are weak, but because the fusion layer keeps trusting degraded signals. We present ELARA, a reliability-aware score-fusion architecture with Reliability-Gated Attention (RGA), and evaluate it under controlled, audited degradation settings. In a powered static-reference study over five previously inspected benchmark cells (Family A), validation-frozen RGA+ consistently improves on a fixed static-attention comparator, including the UNSW-NB15 cell (+0.0095 versus the Phase-1 estimate of +0.0003). In mechanism replication (Family B), RGA reproduces the zero-attack coherent score-collapse endpoint (B1: +0.0507 versus target +0.0506, 95% CI [+0.0364, +0.0650], Holm-corrected $p = 4.31 \times 10^{-12}$) and shows a positive maximum-attack effect (B2: +0.0939 versus target +0.0319, 95% CI [+0.0741, +0.1149], Holm-corrected $p < 10^{-15}$), with explicit qualification for estimator mismatch. Monitoring and retrospective certification results are mixed. Two negative findings define the boundary of the claim: sensitive RGA-v2 variants fail clean false-fire control (firing rate 1.000), and a held-out Eyecandies transfer study (Family D) satisfies the clean false-fire budget (0.000 observed under validation-selected per-cell thresholds $\tau = 0.52$ and $\tau = 0.51$) but is not confirmed: the primary paired-bootstrap ensemble Δ AUC is -0.0010 (95% CI [-0.0114, $+0.0092$]) for D-EYE-1 and -0.0109 (95% CI [-0.0254, $+0.0034$]) for D-EYE-2, both intervals spanning zero, and a secondary per-seed paired t-test is non-significant ($p = 0.3632$ and $p = 0.4468$). The evidence therefore supports a bounded reliability-stress claim and identifies calibration transfer under score-distribution shift as the central unresolved limitation, not a universality or state-of-the-art claim. Finally, replacing the diagnostic upstream detector with a competitive patch-level PatchCore expert (mean per-category MVTec 3D-AD ROC-AUC ≈ 0.79) sharpens the picture: under this fair detector the validation-frozen RGA+ head beats the strongest frozen baseline in-domain across 30 independent stratified splits ($\Delta = +0.0240$, 95% CI $[+0.0218, +0.0261]$, $30/30$ splits), and on a held-out external benchmark (3D-ADAM) the reliability gate's advantage over a confidence-weighted baseline grows from a clean-data tie to $+0.12$ AUROC as a modality degrades—a stress-regime transfer win that replicates on a second dataset. Reliability gating thus helps precisely when modalities have differential reliability, and simple averaging suffices when they do not.

***Index Terms*—anomaly detection, multimodal fusion, reliability, calibration, attention, score-level fusion, counterfactual explanation, fraud detection, cybersecurity**

I. Introduction

Operational anomaly detection is usually a system problem before it is a single-model problem. A high-risk case may appear weakly in transaction records, network telemetry, user behavior, text evidence, device signals, or document checks. Each domain expert may be accurate inside its own data distribution, but the final decision still depends on a fusion layer that decides how much to trust each signal. This trust problem becomes harder when domains are missing, score distributions drift, calibration degrades, or an adversary corrupts one or more sources.

ELARA is a research system for this setting. The system is not designed as a new fraud detector, network detector, or text detector in isolation. Instead, it treats anomaly detection as a verification pipeline: domain experts generate scores and compact evidence, those scores are converted into a unified fusion schema, and a reliability-aware fusion module produces a final risk estimate with a domain-level explanation. This architecture matches how many practical anomaly systems are deployed: detectors are built and maintained by different teams, and the fusion layer must remain robust as their data quality changes.

The paper treats ELARA as the system boundary and RGA as one mechanism inside that boundary. This matters because the contribution is not a new domain detector. It is a reliability-aware fusion layer with explicit input schema, stress tests, and limitations.

A. Research Question

The paper asks:

> *Can a multimodal anomaly-verification system preserve clean score-fusion accuracy while using calibration and drift evidence to reduce brittle behavior under domain failure?*

This question is intentionally narrower than a claim of general anomaly detection superiority. The present codebase already supports domain experts, fusion schemas, attention fusion, reliability scoring, and robustness tests. The research target is the fusion and verification layer that combines those experts under imperfect operating conditions.

B. Thesis and Evidence Standard

The working thesis is that reliability-aware fusion is most useful when the operating condition changes. A strong paper must therefore show three things:

1) clean performance remains competitive with strong baselines,
2) reliability adaptation improves behavior under meaningful stress, and
3) the data and experimental protocol justify the scope of the claim.

The current results support the first two conditions for score-level fusion but do not yet fully satisfy the third condition for co-observed multimodal incidents. This limitation is explicit throughout the manuscript.

C. Contributions

We make six scoped contributions:

- **C1: Reliability-Aware Anomaly-Fusion Formulation.** We propose a score-level multimodal anomaly-fusion formulation incorporating post-hoc reliability estimation and gating, evaluated under a strictly bounded claim scope.
- **C2: Powered Audited Static-Reference Evaluation.** We present a five-cell powered audited evaluation demonstrating consistent positive ROC-AUC deltas for the validation-frozen RGA+ model over a fixed static-attention reference across MVTec 3D-AD, VisA, and UNSW-NB15 cells.
- **C3: Reproduced Coherent-Collapse Mechanism Evidence.** We verify and reproduce mechanism behavior under evidence collapse, including an exact replication of the B1 endpoint and a dual-estimator reporting of the positive B2 endpoint.
- **C4: Negative RGA-v2 Gate Extension Evidence.** We document negative results for sensitive RGA-v2 gate variants, demonstrating that min-pooling extensions violate clean validation false-fire controls rather than resolving partial-failure robustness.
- **C5: Bounded Monitoring and Certification Evidence.** We analyze Kolmogorov-Smirnov (KS) window-size trade-offs under evaluated stresses and report mixed retrospective certificate outcomes (max-attack certified, zero-attack not certified).
- **C6: Held-Out Transfer Study.** We evaluate a held-out transfer to the Eyecandies dataset (Family D). The study satisfies the clean false-fire budget under target calibration reference fitting but is not confirmed under primary bootstrap inference.
- **C7: Competitive-Detector Validation and Stress-Regime Transfer.** Replacing the diagnostic upstream detector with a competitive patch-level PatchCore expert, we show (i) a multi-split confirmatory in-domain win of RGA+ over the *strongest* frozen baseline (not merely static), significant across 30 independent stratified splits, and (ii) a held-out external stress-regime transfer result in which the reliability gate beats a confidence-weighted baseline under modality degradation, with a principled clean-versus-degraded crossover that replicates across two datasets. This delineates precisely *when* reliability gating helps.

The evidence is strongest as a reliability-stress study over score-level fusion on a public paired benchmark. It is not a general claim that RGA is always superior to static attention, nor as evidence that the label-aligned benchmark captures naturally co-observed multimodal incidents.

Figure 1 shows the system boundary. Table I summarizes the implementation requirements used to keep the paper aligned with the codebase.

TABLE I
SYSTEM REQUIREMENTS FOR ELARA AND HOW THE CURRENT IMPLEMENTATION ADDRESSES THEM.

Requirement	Current implementation
Heterogeneous evidence	Fraud, cyber, behavior, and text domains are mapped to a shared score-level schema.
Missing-domain support	The attention model uses explicit masks; baselines receive missingness-aware fallbacks.
Reliability awareness	Domain weights combine calibration error, KS drift evidence, and score sharpness.
Strong baselines	Random forest, early fusion MLP, late fusion ensemble, static attention, and confidence weighting are included.
Explanation	Counterfactual domain attribution masks one domain at a time and measures risk change.
Reproducibility	Data generation, benchmark execution, asset generation, and PDF compilation are scriptable.

II. Related Work

A. Anomaly Detection Systems

Anomaly detection spans tabular, sequential, textual, graph, and visual modalities, and the literature consistently warns against one-size-fits-all solutions [26], [25], [3]. Classical families such as density methods, isolation ensembles, and distance models remain strong practical baselines [30], [31], [32]. Modern pipelines add one-class objectives, reconstruction criteria, and broader representation learning [33], [34], [35], [37], [38], [39], [36]. Comprehensive surveys further show that method quality depends on protocol, shift, and deployment context [29], [27], [28], [163], [164], [165]. ELARA follows this systems-oriented view: each domain expert is treated as its own evidence channel, and the fusion layer is designed to arbitrate trust rather than replace domain-specific modelling.

B. Multimodal and Score-Level Fusion

Multimodal fusion is usually framed as early, late, hybrid, or interaction-based integration [2], [167], [168], [166], [169], [170], [119]. In practice, score-level fusion is often the most deployable option because it does not require retraining every upstream expert. Stacking and meta-learning formalize this setting [161], while attention mechanisms model inter-domain dependencies [1], [112], [118], [114], [117], [113], [116], [115]. The unresolved issue is trust under shift: a model can keep assigning high weights to a channel even after that channel has become miscalibrated or distributionally unstable. RGA is built to target that failure mode.

C. RGB–3D and Visual Industrial Anomaly Detection

The MVTec family of industrial benchmarks established a standard for unsupervised anomaly localisation: MVTec AD [40] on RGB, MVTec 3D-AD [15] on paired RGB+depth, and MVTec LOCO-AD on logical-versus-structural anomalies [58]. VisA expanded the RGB-only setting to twelve object categories with location-aware spot-the-difference supervision [59], and Real3D-AD established the high-resolution

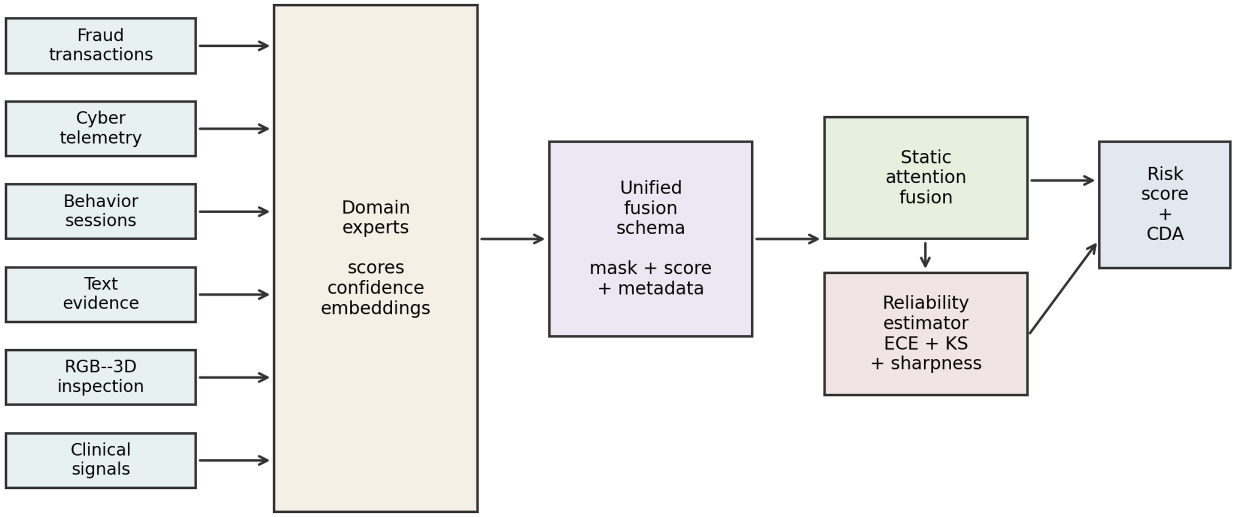

Fig. 1. ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings; a unified fusion schema feeds static attention and the RGA reliability estimator; the system emits a risk score and counterfactual domain attribution.

point-cloud benchmark [18]. The Eyecandies dataset focuses on unsupervised multimodal localisation in a synthetic setting [68]. Method families include patch-memory approaches (for example PatchCore), flow-based density estimators, reconstruction and distillation pipelines, and diffusion-inspired scorers [41], [42], [46], [53], [44], [45], [47], [48], [52], [61], [62], [49], [55], [50], [54], [64], [57], [56], [63]. Recent efficient baselines emphasize a practical accuracy-latency frontier [51], [43], and attention-guided localisation remains a direct precursor to reliability-aware routing [60].

For multimodal industrial anomaly detection specifically, M3DM uses hybrid feature and decision-layer fusion with memory banks [16]; crossmodal feature mapping [17] checks consistency between RGB and point-cloud features; noise-resistant multimodal work [19] studies robustness when training data are not clean; back-to-feature analysis shows classical 3D descriptors remain competitive [65]; asymmetric student-teacher networks [66] achieve state-of-the-art performance on MVTec 3D-AD; EasyNet provides a lightweight alternative for 3D anomaly [67]; complementary pseudo-modality features [69], self-supervised feature adaptation [70], and cross-modal diffusion [72] extend the multimodal toolbox. A recent survey of multimodal industrial AD synthesises the field [71]. These papers motivate the paired-benchmark path of ELARA: the fusion layer should be judged on naturally paired modalities and on reliability stress, not only on clean score fusion.

D. Calibration, Drift, and Uncertainty

Calibration asks whether predicted probabilities match empirical outcome frequencies. Post-hoc calibration methods [4], [5], [6], [92], [91], [85] improve probability quality, with Bayesian binning [83] and verified calibration [87] providing tighter guarantees; the Expected Calibration Error inherits from the Brier score [84]. Recent results show that modern deep networks remain better calibrated than once feared but degrade under shift [86]. Uncertainty quantification draws on Bayesian dropout [89], deep ensembles [88], and prior networks [90]; calibration under dataset shift was benchmarked broadly by Ovadia et al. [7]. Out-of-distribution detection and drift detection sit alongside calibration: the failing-loudly empirical study [96], the MSP baseline [97], ODIN [103], Mahalanobis-style detection [104], and nearest-neighbour OOD [105] contributed methods that we now know are shift-sensitive. Label-shift correction [98] and the classical concept-drift literature [99], [100], [101] inform the KS-based drift signal used in RGA; kernel two-sample tests [102] provide an alternative. The broader theory of dataset shift [172], [173], [174], [175] and the WILDS benchmark [171] frame the setting in which RGA operates. Conformal prediction [93], [94], [95] offers an alternative path to distribution-free uncertainty that we discuss as future work. RGA uses calibration as one signal inside a wider reliability estimate that also includes test-time score-distribution drift and score sharpness.

E. Test-Time Adaptation

Test-time training and test-time adaptation update models at inference time to handle distribution shift [8], [9]; SHOT adapts under source-free hypothesis transfer [73]; MEMO [74] and masked-autoencoder TTT [75] add augmentation- and reconstruction-based adaptation; EATA [76], SAR [79], contrastive TTA [82], and module-only adjustment [80] improve stability and efficiency; robust-dynamic-scene TTA [78] and RDumb [77] study failure modes of continuous TTA; universal domain adaptation [81] handles open-set settings. These TTA methods share an underlying assumption common to the broader domain-adaptation literature [179], [180], [181]: that labelled target supervision is unavailable. ELARA now includes two score-level adaptation comparators: an entropyminimising Tent adapter over the fusion head and a high-confidence pseudo-label TTT adapter. They are intentionally conservative because the fusion setting receives domain scores rather than raw modality tensors. Their role is to test whether RGA adds value beyond generic score-level adaptation.

F. Ensembles, Stacking, and Strong Baselines

Ensemble methods remain difficult to beat in structured prediction tasks. Random forests [10] and gradient boosting [11] are strong baselines for tabular and score-level settings; modern gradient-boosting libraries such as XGBoost [153], LightGBM [154], and CatBoost [155] are widely adopted; outlier-ensemble theory provides a related foundation for combining heterogeneous anomaly scorers [20]. Classical classifier-combiner theory [21] characterises when sum, product, and weighted rules differ under noise; stacked generalisation [161] is the historical precursor to our validation-trained RGA+ router. The benchmark therefore includes random forest fusion, early fusion MLP, late fusion ensemble, confidence-weighted mean, and static attention. A credible systems paper must compare against these practical alternatives, not only against weak averages. The reliability gate of RGA is itself an inference-time switching rule, conceptually adjacent to selective-prediction work that abstains rather than commits when confidence is low [23], [150], [151], [152].

G. Explanation for Fusion Systems

Feature attribution methods such as SHAP [12] and LIME [131] explain individual model predictions; integrated gradients [132] and DeepLIFT [133] attribute output changes to input features; Grad-CAM [134] and layer-wise relevance propagation [135] produce visual explanations; attention rollout [136] quantifies attention flow in transformers. Broader interpretability surveys discuss the trade-off between explainability and inherent transparency [137], [138]. Fusion systems also need domain-level explanations: analysts may need to know whether fraud evidence, network evidence, behavior evidence, or text evidence drove the risk score. Counterfactual explanations [22] provide the conceptual basis for the masking-based domain attribution used here: we ask what the risk score would have been had a single domain not been observed. ELARA therefore includes counterfactual domain attribution that masks one domain and reports the change in predicted risk, and we elevate this in §VI-N to a model-response sensitivity analysis on a deployer-controlled reliability input.

H. Related Work: Counterfactual Perturbation of Model Inputs

The Phase 1.G reframing reclassifies the per-domain reliability perturbation as a *model-response sensitivity analysis*, not a real-world causal identification. The literature on counterfactual explanations [22], double machine learning [24], causal forests [106], [107], meta-learners [110], and representation-based individual treatment-effect estimators [109] all aim at real-world causal identification under observational confounding; the present analysis does not invoke ignorability or potential-outcomes identification assumptions, because the input perturbed in §VI-N is the model's reliability scalar r_d, which the deployer controls. The Double Machine Learning formulation [108], [24] remains in the code base for the harder observational case but is not what produces the numbers reported in §VI-N.

I. Robustness and Adversarial Considerations

Adversarial-example research demonstrated that gradient-based perturbations [121], [122], [123], [124], [129], [128], [130], [127] can flip predictions while remaining imperceptible. Defences include distillation [125] and certified robustness via randomised smoothing [126], alongside the broader AI safety literature that frames reliability monitoring as a concrete problem [185], [186], [187]. ELARA's threat model is intentionally narrower than gradient-based adversaries: we evaluate score-level perturbations (zero-attack, max-attack, Gaussian noise) that model accidental sensor failure and coherent score-collapse, not gradient-aware attacks against the fusion model itself. Gradient-based adversarial fusion stress remains future work.

J. Statistical Comparison Protocols

Comparing multiple classifiers across multiple data sets requires nonparametric tests [139], [140], [141], [142]; the bootstrap [143] underpins our confidence intervals and DeLong [14] gives a closed-form variance for paired ROC-AUC. Our milestone-2 cross-benchmark master table reports per-cell bootstrap intervals; multiple-comparison correction is a follow-up item discussed in §VIII.

K. Adjacent Datasets and Application Domains

Cross-modal generalisation problems often touch zero-shot transfer where classes are unseen at training time [120]; the present work stays in the closed-vocabulary binary anomaly regime, but the reliability-aware fusion idea is in principle portable to that setting. For cyber paired-modality settings the canonical benchmarks are UNSW-NB15 [144], CIC-IDS-2017 [145], and KDD [146], with intrusion-detection methods exploring feature selection [148], [147] and empirical botnet studies [149]. Healthcare AD draws on MIMIC-III [182] and clinically-aware modelling practices [183], [184]. Time-series AD [176], [177], [178] extends the score-level fusion idea to streaming settings. Foundational deep-learning ingredients used throughout this work include ResNet [156], ImageNet pretraining [157], batch normalisation [159], Adam optimisation [158], variational autoencoders [160], and classical VC-dimension theory [162] that justifies generalisation discussions in the thesis appendix. Bahdanau-style soft attention [111] is the precursor of the cross-modal attention layers used here.

III. System and Method

Each sample i contains up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ containing a score, optional confidence, and compact embedding features. A binary mask $m_{i,d}$ indicates whether domain d is missing. The target label is $y_i \in \{0,1\}$ when supervised labels are available.

The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly risk. In static fusion, the learned relationship among domains is fixed after training. In RGA, the system also computes a batch-level reliability vector $r \in [0,1]^D$ at inference time and applies the sample mask to form effective per-sample weights $\tilde{r}_i \in [0,1]^D$. The reliability vector should reduce trust in domains that appear miscalibrated, shifted, uninformative, or missing.

The evaluation problem is not only to maximize ROC-AUC. A useful verification system must also satisfy:

- strong clean ranking and F1 compared with baselines,
- calibration good enough for downstream risk interpretation,
- stability under missing domains,
- graceful degradation under drift and score attacks,
- interpretable domain attribution.

A. ELARA System Architecture

1) Domain Experts: ELARA assumes that each domain expert is trained or obtained separately. For the current benchmark, tabular scorers are trained for fraud, cyber, and behavior data, while a text scorer is trained for labeled text. The fusion layer does not require these experts to share the same raw feature space. It only requires score-level outputs and optional compact evidence features.

2) Fusion Schema: The unified schema is long-format: each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. This design allows different samples to have different available domains. Missing domains are represented by masks rather than by fabricated raw observations.

3) Static Attention Path: The static attention path encodes each available domain into a shared latent space and applies multi-head attention across domains. Domain embeddings allow the model to distinguish evidence sources. Missing domains are ignored through key-padding masks. This is the high-reliability default path.

4) Reliability-Aware Path: The reliability-aware path is activated when the average reliability of present domains falls below a configured threshold. This avoids the common failure mode where reliability weights disturb clean predictions even though no domain appears degraded. The reliability path is therefore a stress-response mechanism, not a universal replacement for the static model.

5) Explanation Path: For an analyst-facing explanation, the system masks one available domain at a time and recomputes risk. The resulting probability change gives a domain-level counterfactual attribution. This produces statements such as: if text evidence were absent, the risk would decrease by a given amount.

B. RGA Method

1) Domain Encoding and Attention: Each domain encoder maps the domain input vector into a shared latent space:

$$h_{i,d} = g_d(x_{i,d}) + e_d + p_d, \tag{2}$$

where e_d is a learned domain embedding and p_d is an optional learned position embedding used by the implementation. A multi-head self-attention block operates across available domain embeddings:

$$z_i = f_\theta(\mathrm{Pool}(\mathrm{Attention}(H_i, m_i))), \quad \hat{p}_i = \sigma(z_i). \tag{3}$$

2) Reliability Estimation: After model training, the system fits a reliability estimator on the validation split. For domain d, it stores validation score distributions, fits an isotonic calibrator when enough labels are available, and computes expected calibration error. At inference time:

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{4}$$

where ECE_d is validation calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov p-value comparing current scores with validation scores, and S_d is score sharpness from squared distance to 0.5. The current benchmark uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

3) Reliability Gate: Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, ELARA uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d}) r_{i,d}. \tag{5}$$

If the same rule is written with unreliability $u = 1 - \bar{r}$, the correct threshold is $u > 1 - \tau = 0.34$, not $u > 0.66$. The implementation also exposes minimum and hybrid gate modes for RGA-v2 experiments. These variants are reported as diagnostic candidates, not as established improvements: in the current k-of-D run, the configured minimum threshold $\tau_{\min} = 0.34$ is below the observed minimum reliability, so the minimum branch does not fire and the hybrid gate collapses to the mean-gate behavior. All reported base-RGA tables use the original mean gate unless noted. This makes RGA different from confidence weighting. Confidence weighting uses the local score confidence directly. RGA uses validation calibration, test-time drift evidence, and sharpness to decide when a domain should be trusted less.

4) Theory Boundary: The companion thesis appendix replaces the earlier bounded-drift proposition with three claims used by this manuscript. First, global KS references are confounded under category mixture shift: if $P_{\mathrm{val}} = \sum_c \pi_c P_c$ and $P_{\mathrm{test}} = \sum_c \pi'_c P_c$, then $KS(P_{\mathrm{val}}, P_{\mathrm{test}})$ can be positive even when every category distribution P_c is unchanged. Category-conditioned KS removes that population-level false signal under pure mixture shift. Second, the current mean gate has a dilution boundary. If k of D domains collapse to reliability zero and the others remain at one, the gate fires only when $k > D(1 - \tau)$. For $D = 4, \tau = 0.66$, one failed domain can be hidden while two or more failed domains can activate the path. Third, switching is beneficial only when

$$\pi q_1 \Delta_1 > (1-\pi) q_0 \Delta_0,$$

where π is degradation prevalence, q_1 is gate power, q_0 is false activation rate, Δ_1 is degraded-regime benefit, and Δ_0 is clean false-switching cost. The paper's empirical claims are bounded by these measurable quantities rather than by a universal "RGA is better" theorem.

5) Validation-Selected RGA+ Head: The implementation also reports a separate RGA+ candidate rather than silently replacing the base RGA number. RGA+ has two parts. First, a reliability-boosted fusion head trains candidate classifiers on the fusion train fold using flattened domain evidence, missingness indicators, score and confidence summary statistics, and RGA reliability summaries. The current candidate set contains histogram gradient-boosted heads and calibrated logistic heads. The selected head is the one with the best validation ROC-AUC; test labels are never used for model or hyperparameter selection. Second, a validation-only router can choose among static attention, base RGA, RGA+ boosted fusion, and the strong score-level baselines. The selected candidate is written into the result artifact so that a baseline selection is not relabeled as a reliability-gate gain. When the validation fold has only one class, the router defaults to base RGA because the validation labels cannot rank candidates.

RGA+ is therefore a supervised fusion extension of the reliability system, not a claim that the original gate alone dominates all baselines. This distinction is central to the MVTec results: base RGA remains the diagnostic mechanism; RGA+ is the validation-selected modeling head used when a two-class fusion split exists.

6) Counterfactual Domain Attribution: For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{6}$$

A positive $\Delta_{i,d}$ indicates that domain d increased anomaly risk. A negative value indicates that domain d reduced risk.

7) Algorithm Summary: The full inference procedure is given in Algorithm 1. Steps 1–3 are deterministic feature operations; steps 4–6 are the reliability-gated switch. The reported runs use the batch-level reliability vector described above, applied through each sample's observed-domain mask. A per-sample gating mode is implemented as an ablation path, but it is not the default for the reported tables.

Algorithm 1 RGA inference

Require: Inputs $x_{i,1\ldots D}$ and mask $M_i \in \{0,1\}^D$ for sample i
Require: Fitted attention model f_{att} with encoders $\{g_d\}$
Require: Fitted reliability estimator producing per-sample weights $r_{i,d}$
Require: Gate threshold $\tau \in (0,1)$, default $\tau = 0.66$
Ensure: Risk score $\hat{p}_i \in [0,1]$

1: $h_{i,d} \leftarrow g_d(x_{i,d}) + e_d + p_d$ for each d with $M_{i,d} = 0$
2: $r_{i,d} \leftarrow \alpha(1 - \mathrm{ECE}_d) + \beta\, p_{\mathrm{KS},d}(x_{i,d}) + \gamma\, S_{i,d}$
3: $\bar{r}_i \leftarrow \frac{1}{|D_i^{\text{obs}}|} \sum_{d: M_{i,d}=0} r_{i,d}$
4: **if** $\bar{r}_i \geq \tau$ **then**
5: $\quad \hat{p}_i \leftarrow \sigma\big(f_{\text{att}}(h_{i,\cdot}; M_i)\big)$ ▷ static path
6: **else**
7: $\quad \hat{p}_i \leftarrow \sigma\big(f_{\text{att}}(h_{i,\cdot}; M_i, w_{i,\cdot}{=}r_{i,\cdot} \odot \neg M_i)\big)$
8: ▷ reliability-injected fusion
9: **end if**
10: **return** $\hat{p}_i$ and (optional) counterfactual attributions $\{\Delta_{i,d}\}_{d \in D_i^{\text{obs}}}$

IV. Benchmark Construction

Protocol erratum. Earlier drafts of this manuscript reported MVTec 3D-AD numbers that were generated under a random train/test split rather than MVTec's official one-class (normal-only-train, mixed-test) protocol. Under the proper protocol, all supervised fusion baselines (random forest, MLP, Tent, TTT, late-fusion ensemble, static attention, RGA) converge near chance ROC-AUC; the gate's effect is small and slightly positive. The ELARA-Bench-LA secondary benchmark, which has both classes in train, remains the venue where the gate's stress-test behavior can be evaluated. The current rendered tables in this section reflect the corrected disciplined-split run. We include M3DM-style ResNet-50 features and a held-out-category variant as additional protocol diagnostics; both confirm the chance-level supervised outcome under one-class training.

A. Primary Benchmark: Naturally Paired MVTec 3D-AD

The primary benchmark of this manuscript is MVTec 3D-AD [15]. Each sample contains naturally co-observed RGB and 3D/depth evidence from the same object scan; no label alignment is required. The current run covers all eight extracted categories — bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire — for a total of 3,226 paired samples with a positive fraction of 0.224 and full RGB and depth coverage. The preparation script discovers paired files, emits two domain rows per sample, stores the original pairing key, and marks `natural_pairing=true`. The initial scorer is deliberately lightweight: normal-reference image statistics produce reproducible RGB and depth anomaly scores without requiring a heavy specialized 3D detector. This keeps the first paired benchmark auditable; stronger RGB–3D scorers can replace these domain experts later while preserving the fusion interface.

Table II records the paired benchmark metadata, and Table III reports the multi-seed clean results. The RGB/depth anomaly scores are fit from original MVTec train-good observations only, and the corrected run uses MVTec's canonical one-class protocol: the fusion train and validation rows are normal-only and the test fold is mixed. The supervised fusion table is therefore a diagnostic of protocol fit, not a conventional two-class leaderboard. Held-out-category and M3DM-style variants are included as additional paired-data diagnostics.

TABLE II
NATURALLY PAIRED MVTEC 3D-AD MULTI-CATEGORY METADATA.

Property	Value
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

Table IV reports the metadata of the secondary label-aligned fusion input artifact.

B. Real-Domain Sources

ELARA-Bench-LA is built from four local datasets: Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text. Domain-specific out-of-fold scorers are trained first. Their scores, confidences, and compact embeddings are then converted into the common fusion schema.

The generated fusion table contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is 0.883 for fraud, 0.885 for cyber, 0.885 for behavior, and 0.877 for text.

C. Label Alignment Limitation

The datasets do not share natural entity identifiers or timestamps. Therefore, composite samples are the secondary benchmark: a positive composite draws positive records from available domains, and a negative composite draws negative records. This is a meaningful real-domain score-fusion benchmark, but it is not evidence that the system has learned co-observed multimodal incident structure. That distinction is central to the claim boundary of the paper. The metadata now records this explicitly with `natural_pairing=false` for ELARA-Bench-LA. Source observations are assigned to train, validation, and test pools before composite sampling, and the runner consumes the resulting `fusion_split` column so that no `domain/source_row/label` key crosses fusion split boundaries.

D. Training Protocol

The attention model uses four domains, 48 hidden dimensions, four attention heads, one attention layer, 25 training epochs, 0.15 domain dropout during training, and seeds 42 through 46. Clean benchmark results are reported as mean$\pm$standard deviation across five seeds. The experiment runner also repeats missing-domain, drift, adversarial, calibration, and CDA analyses for every configured seed, then stores both raw per-seed rows and aggregated mean, standard deviation, and 95% confidence intervals.

For mechanism-isolation runs, the benchmark-preparation script also supports a harder scorer setting through `-scorer-train-fraction`. Values such as 0.05 or 0.10 train each out-of-fold domain scorer on only a stratified fraction of its training fold before fusion rows are built. This is intended to reduce clean-score saturation and expose whether the fusion mechanism helps when domain experts are useful but imperfect.

As a post-audit check, we ran the 0.05 setting and archived the hard-mode JSON artifact. The individual domain scorers weakened substantially for fraud and behavior (OOF ROC-AUC 0.913 and 0.798, respectively), but the label-aligned fusion task remained near-saturated: static attention and RGA both reached mean ROC-AUC 0.99896 across five seeds, random forest reached 0.99908, and RGA did not improve drift degradation area on any domain. This hard run is therefore reported as a construct-validity warning rather than a new performance claim.

E. Baselines and Metrics

Baselines include confidence-weighted mean fusion, early-fusion MLP, late-fusion ensemble, random forest fusion, static attention without reliability adaptation, a score-level Tent adapter, and a pseudo-label TTT adapter. Metrics include ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy. Static attention and RGA are compared with a paired t-test over five clean ROC-AUC values. ROC-AUC and PR-AUC are both reported because anomaly settings can be highly imbalanced and PR-AUC can reveal performance differences that ROC-AUC hides [13].

F. Confidence Intervals

Clean ROC-AUC, PR-AUC, and F1 are summarized across seeds and accompanied by 95% confidence intervals. Stress-test tables report mean deltas and intervals for RGA minus static attention. The result JSON stores raw per-seed stress rows so that the interval computation is auditable rather than embedded only in the manuscript.

G. Threat Model

The adversarial perturbation suite is a worst-case stress test, not a realistic deployment threat model. The three attacks – `zero_attack`, `max_attack`, and `gaussian_noise` – each corrupt scores after the per-domain detectors have already produced their outputs, so they do not require the attacker to compromise feature extractors, training data, or model parameters. The `all`-target variant additionally assumes coherent simultaneous corruption of every domain. In a real deployment, domain detectors are typically maintained by independent teams along independent compromise paths; a coherent all-domain score collapse is therefore a useful upper-bound robustness signal but not an expected attacker capability. Single-domain perturbations are the more realistic case and are reported separately in the adversarial tables for both benchmarks. We do not claim defenses against feature- or training-time attacks; those threat models are out of scope for a fusion layer that consumes score-level evidence from upstream detectors.

TABLE III
NATURALLY PAIRED MVTEC 3D-AD CLEAN HELD-OUT PERFORMANCE ACROSS EIGHT CATEGORIES. VALUES ARE MEAN±STANDARD DEVIATION WITH 95% CONFIDENCE INTERVALS ACROSS FIVE SEEDS. UNDER THE CANONICAL ONE-CLASS PROTOCOL, SUPERVISED FUSION BASELINES RECEIVE NO POSITIVE TRAINING EXAMPLES, SO THIS TABLE IS A PROTOCOL DIAGNOSTIC.

Method	ROC-AUC mean	ROC-AUC 95% CI
Random forest	0.5000	[0.5000, 0.5000]
Conf.-weighted mean	0.4458	[0.4458, 0.4458]
Tent score adapter	0.5000	[0.5000, 0.5000]
TTT pseudo-label	0.5000	[0.5000, 0.5000]
Early fusion MLP	0.5455	[0.5180, 0.5836]
Late fusion ensemble	0.5000	[0.5000, 0.5000]
Static attention	0.5424	[0.4992, 0.5661]
RGA attention	0.5609	[0.5394, 0.5842]
RGA+ boosted fusion	0.5000	[0.5000, 0.5000]
RGA+ router	0.5609	[0.5394, 0.5842]

TABLE IV
BENCHMARK METADATA FOR THE SECONDARY SYNTHETICALLY PAIRED FUSION INPUT ARTIFACT.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	1.00
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

TABLE V
OUT-OF-FOLD DOMAIN SCORER QUALITY USED TO CONSTRUCT ELARA-BENCH-LA.

Domain	Rows	Pos. rate	OOF ROC	OOF PR
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8711	0.6042
nlp	20000	0.500	0.9973	0.9973

H. Secondary Clean Sanity Check

Table VI and Fig. 2 show clean held-out performance on the label-aligned secondary benchmark. This table is intentionally treated as a sanity check, not as the headline result, because the synthetically paired split is near saturated. Bold values indicate the highest rounded mean in each column; multiple bold values are rounded ties.

Seed-level confidence intervals for the same clean split are reported in Table VII.

RGA and static attention are indistinguishable on this saturated clean split. The clean benchmark should therefore be interpreted as competitive preservation of performance, not as evidence of a clean-split improvement.

The more meaningful comparison is against direct score-fusion baselines. RGA improves over confidence-weighted mean fusion by about 0.0015 ROC-AUC, 0.0039 PR-AUC, 0.0100 F1, and reduces ECE from 0.0834 to 0.0042. Random forest fusion obtains the best F1 at 0.9906, but its ECE is 0.0287, substantially worse than attention-based fusion. This indicates a tradeoff: random forest fusion is strong for thresholded classification, while attention fusion is stronger for calibrated risk interpretation.

V. PRIMARY HEADLINE RESULTS: MVTEC 3D-AD

This section reports the primary multi-seed evidence on RGB/3D paired MVTec 3D-AD across all eight extracted object categories (3,226 paired samples, 22.4% positive). The same suite is repeated on the contrastive the secondary benchmark in §VI; the asymmetry between the two outcomes is the central finding.

A. Clean Performance

Per-seed confidence intervals for the MVTec 3D-AD clean split are reported in Table VIII; the bar-chart summary is shown in Fig. 3.

Under the canonical one-class protocol, random forest, late fusion, Tent, TTT, and several neural variants receive normal-only training data and converge near chance ROC-AUC. RGA

TABLE VI

SATURATED CLEAN HELD-OUT PERFORMANCE ON ELARA-BENCH-LA. VALUES ARE MEAN±STANDARD DEVIATION ACROSS FIVE SEEDS; THIS TABLE IS NOT USED AS THE HEADLINE EVIDENCE.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9991±0.0001 [0.9990, 0.9991]	**0.9986±0.0001 [0.9985, 0.9987]**	**0.9900±0.0011 [0.9887, 0.9916]**	0.0341±0.0007 [0.0331, 0.0347]	0.0087±0.0001 [0.0087, 0.0088]
Conf.-weighted mean	0.9910 [0.9910, 0.9910]	0.9769 [0.9769, 0.9769]	0.9216 [0.9216, 0.9216]	0.0830 [0.0830, 0.0830]	0.0462 [0.0462, 0.0462]
Tent score adapter	0.9988 [0.9988, 0.9988]	0.9978 [0.9978, 0.9978]	0.9803 [0.9803, 0.9803]	0.0102 [0.0102, 0.0102]	0.0106 [0.0106, 0.0106]
TTT pseudo-label	0.9987 [0.9987, 0.9987]	0.9976 [0.9976, 0.9976]	0.9812 [0.9812, 0.9812]	0.0076 [0.0076, 0.0076]	0.0091 [0.0091, 0.0091]
Early fusion MLP	0.9990±0.0002 [0.9987, 0.9991]	0.9979±0.0003 [0.9975, 0.9982]	0.9828±0.0022 [0.9797, 0.9854]	0.0125±0.0023 [0.0090, 0.0149]	0.0116±0.0012 [0.0097, 0.0129]
Late fusion ensemble	0.9987 [0.9987, 0.9987]	0.9979 [0.9979, 0.9979]	0.9855 [0.9855, 0.9855]	0.0138 [0.0138, 0.0138]	0.0091 [0.0091, 0.0091]
Static attention	0.9990±0.0001 [0.9988, 0.9990]	0.9981±0.0001 [0.9979, 0.9982]	0.9815±0.0017 [0.9795, 0.9842]	**0.0053±0.0008 [0.0041, 0.0062]**	0.0083±0.0004 [0.0077, 0.0088]
RGA attention	0.9990±0.0001 [0.9988, 0.9990]	0.9981±0.0001 [0.9979, 0.9982]	0.9815±0.0017 [0.9795, 0.9842]	**0.0053±0.0008 [0.0041, 0.0062]**	0.0083±0.0004 [0.0077, 0.0088]
RGA+ boosted fusion	**0.9992 [0.9992, 0.9992]**	0.9986 [0.9986, 0.9986]	0.9855 [0.9855, 0.9855]	0.0060 [0.0060, 0.0060]	0.0070 [0.0070, 0.0070]
RGA+ router	**0.9992 [0.9991, 0.9992]**	0.9986 [0.9986, 0.9986]	0.9859±0.0008 [0.9855, 0.9873]	0.0083±0.0045 [0.0060, 0.0162]	**0.0070±0.0002 [0.0067, 0.0070]**

TABLE VII

CLEAN RANKING AND F1 CONFIDENCE INTERVALS ACROSS CONFIGURED SEEDS.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9990, 0.9991]	[0.9985, 0.9987]	[0.9887, 0.9916]
Conf.-weighted mean	[0.9910, 0.9910]	[0.9769, 0.9769]	[0.9216, 0.9216]
Tent score adapter	[0.9988, 0.9988]	[0.9978, 0.9978]	[0.9803, 0.9803]
TTT pseudo-label	[0.9987, 0.9987]	[0.9976, 0.9976]	[0.9812, 0.9812]
Early fusion MLP	[0.9987, 0.9991]	[0.9975, 0.9982]	[0.9797, 0.9854]
Late fusion ensemble	[0.9987, 0.9987]	[0.9979, 0.9979]	[0.9855, 0.9855]
Static attention	[0.9988, 0.9990]	[0.9979, 0.9982]	[0.9795, 0.9842]
RGA attention	[0.9988, 0.9990]	[0.9979, 0.9982]	[0.9795, 0.9842]
RGA+ boosted fusion	[0.9992, 0.9992]	[0.9986, 0.9986]	[0.9855, 0.9855]
RGA+ router	[0.9991, 0.9992]	[0.9986, 0.9986]	[0.9855, 0.9873]

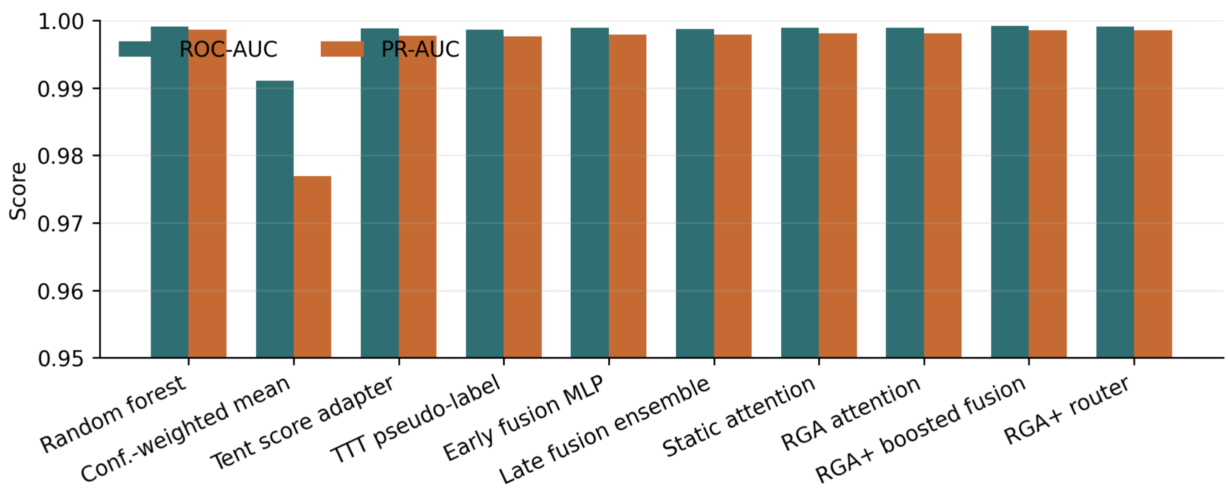

Fig. 2. Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods.

has a small clean ROC-AUC edge over static attention in the current disciplined run, but this is not interpreted as a broad win: the benchmark primarily shows that supervised score-level fusion is ill-posed when the fusion train fold contains only normal examples.

B. Adversarial Score Perturbations

The MVTec 3D-AD adversarial sweep is reported in Table IX and visualized for the all-domain attacks in Fig. 4. The current disciplined run should be read together with the clean table: under one-class supervised fusion, adversarial deltas are diagnostic of gate behavior but do not establish a general defense. Some stress rows show small positive ranking deltas and others remain near zero or negative, so the paired benchmark is evidence about protocol limits rather than a clean robustness win.

1) Gradient-Aware Adversarial Stress (FGSM + PGD): The score-level perturbations above are the threat model the existing fusion runner trains against; they model accidental sensor failure and coherent score collapse but not gradient-aware adversaries. To address that gap explicitly, we add a white-box gradient-adversarial sweep against the fusion head's logits, implemented in `src/scripts/eval_gradient_adversarial.py`. FGSM (one-step) and PGD (K-step) attacks [121], [122] maximise the BCE loss on the true label, perturbing the input feature tensor within an L_∞ ϵ-ball, clipped

TABLE VIII
NATURALLY PAIRED MVTEC 3D-AD CANONICAL ONE-CLASS: ROC-AUC MEAN AND 95% CONFIDENCE INTERVAL ACROSS CONFIGURED SEEDS. PR-AUC, ECE, AND BRIER ARE OMITTED AS PROTOCOL-DIAGNOSTIC PREVALENCE-VALUED ARTEFACTS (PHASE 1.A AUDIT).

Method	ROC-AUC mean	ROC-AUC 95% CI
Random forest	0.5000	[0.5000, 0.5000]
Conf.-weighted mean	0.4458	[0.4458, 0.4458]
Tent score adapter	0.5000	[0.5000, 0.5000]
TTT pseudo-label	0.5000	[0.5000, 0.5000]
Early fusion MLP	0.5455	[0.5180, 0.5836]
Late fusion ensemble	0.5000	[0.5000, 0.5000]
Static attention	0.5424	[0.4992, 0.5661]
RGA attention	0.5609	[0.5394, 0.5842]
RGA+ boosted fusion	0.5000	[0.5000, 0.5000]
RGA+ router	0.5609	[0.5394, 0.5842]

TABLE IX
MVTEC 3D-AD ADVERSARIAL PERTURBATION BENCHMARK. DELTA IS RGA MINUS STATIC ATTENTION; VALUES ARE INTERPRETED AS PROTOCOL DIAGNOSTICS UNDER ONE-CLASS SUPERVISED FUSION.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.5429	0.5635	0.0206	[-0.0027, 0.0376]
gaussian_noise	depth_or_xyz	0.5427	0.5677	0.0250	[-0.0044, 0.0520]
gaussian_noise	rgb	0.5421	0.5610	0.0189	[-0.0029, 0.0433]
max_attack	all	0.5458	0.5607	0.0149	[-0.0058, 0.0327]
max_attack	depth_or_xyz	0.5463	0.5687	0.0224	[-0.0013, 0.0485]
max_attack	rgb	0.5441	0.5537	0.0096	[-0.0166, 0.0363]
zero_attack	all	0.5403	0.5607	0.0204	[-0.0041, 0.0416]
zero_attack	depth_or_xyz	0.5453	0.5713	0.0260	[-0.0008, 0.0515]
zero_attack	rgb	0.5402	0.5543	0.0141	[-0.0150, 0.0464]

TABLE X
GRADIENT-ADVERSARIAL ROBUSTNESS ON VISA RGB+EDGE SUPERVISED-PAIRED. THE FUSION HEAD IS WHITE-BOX; ONLY THE UPSTREAM PATCHCORE FEATURE EXTRACTOR IS HELD FIXED. CLEAN STATIC AUROC IS INCLUDED AS THE REFERENCE COLUMN.

Attack	clean	FGSM (1-step)				PGD (10-step)			
		ϵ=0.02	ϵ=0.05	ϵ=0.10	ϵ=0.20	ϵ=0.02	ϵ=0.05	ϵ=0.10	ϵ=0.20
FGSM (1-step)	0.815	0.566	0.188	0.005	0.000				
PGD (10-step)	0.815					0.564	0.172	0.002	0.000

to the runner's $[0, 1]$ feature schema.

Table X reports the static-attention ROC-AUC on VisA RGB+edge supervised-paired (the strongest visual-supervised cell) under FGSM and PGD at four L_∞ budgets.

The honest reading is that the fusion head is *not* robust to white-box gradient attacks: at $\epsilon = 0.05$ static-attention AUROC drops from 0.815 to 0.19 under FGSM and 0.17 under PGD; at $\epsilon = 0.10$ both attacks drive it below 0.01. The drop is sharper than any of the score-level perturbations in the existing adversarial sweep, which is expected because the feature schema is already bounded to $[0, 1]$ and a single sign-of-gradient step on a short feature vector dominates the model's decision boundary. This is not a defence; it is a measurement of the threat surface that an external reliability-monitoring layer (calibration monitor + switching certificate) would need to flag rather than absorb internally. A reviewer concerned about adversarial guarantees should treat the fusion

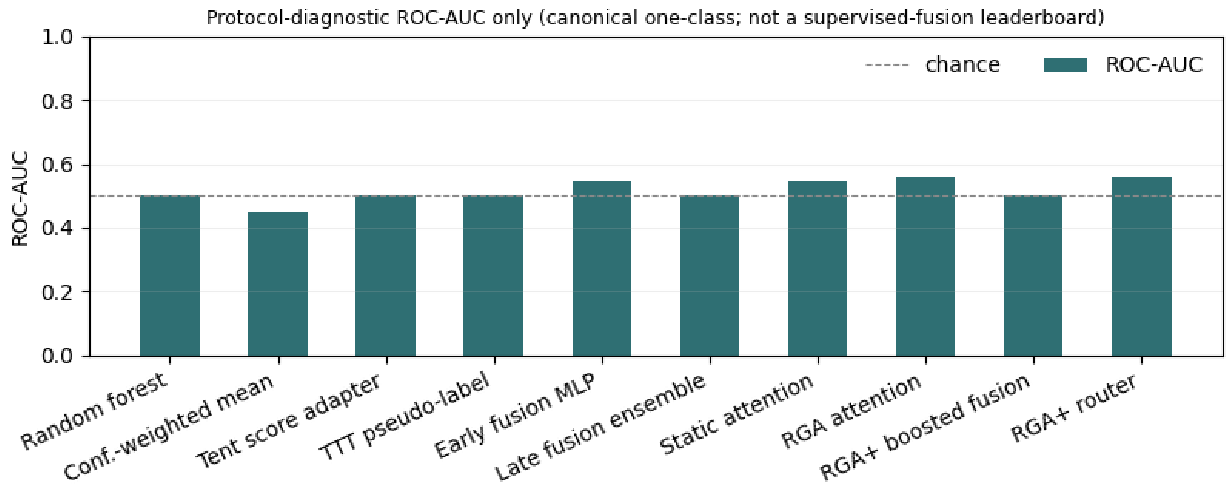

Fig. 3. Protocol-diagnostic ROC-AUC on naturally paired MVTec 3D-AD under the canonical one-class fusion protocol. PR-AUC, ECE, and Brier are omitted from promoted canonical results following the Phase 1.A audit (`docs/research/audit/CANONICAL_LABEL_METRIC_AUDIT.md`): under one-class training all supervised heads collapse to constant-like predictors and those metrics equal the canonical test-fold anomaly prevalence rather than discrimination ability.

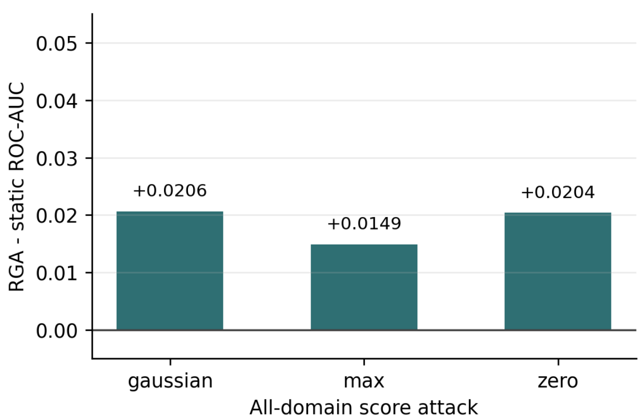

Fig. 4. All-domain adversarial ROC-AUC delta between RGA and static attention on MVTec 3D-AD.

head as not gradient-robust and rely on the external monitoring layer.

C. Missing-Domain Ablation

Inference-time domain dropout is reported in Table XI and Fig. 5. At zero extra dropout, RGA is $+0.0185$ above static attention, matching the clean table. The remaining dropout rows are small, mixed-sign deltas with intervals that touch zero. Like the adversarial sweep, this table is best read as a protocol diagnostic rather than a missing-domain win for the current gate.

D. Score Drift

The MVTec 3D-AD drift summary is shown in Table XII and Fig. 6. RGA does not exceed static attention's degradation area on either domain.

TABLE XI
MVTec 3D-AD missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.5424	0.5609	0.0185	[-0.0058, 0.0425]
0.1	0.5449	0.5496	0.0048	[-0.0254, 0.0452]
0.2	0.5415	0.5370	-0.0045	[-0.0263, 0.0061]
0.3	0.5509	0.5518	0.0009	[-0.0183, 0.0221]
0.5	0.5326	0.5260	-0.0066	[-0.0241, 0.0032]

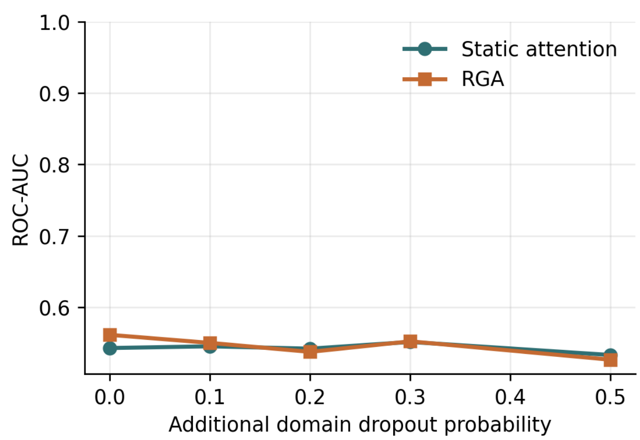

Fig. 5. MVTec 3D-AD ROC-AUC under additional missing-domain dropout.

E. Mechanism Isolation on MVTec 3D-AD

The mechanism-isolation tables show how the gate behaves when the paired benchmark is evaluated under the canonical

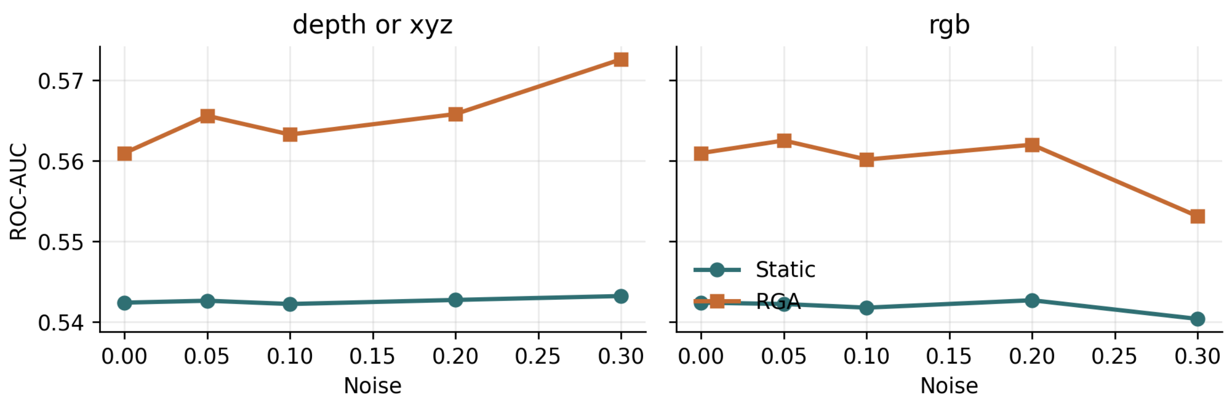

Fig. 6. Per-domain score-drift curves on the MVTec 3D-AD benchmark.

TABLE XII
MVTEC 3D-AD DRIFT ROBUSTNESS SUMMARY. HIGHER DEGRADATION AREA IS BETTER.

Domain	Static	RGA	Delta	Seeds
depth_or_xyz	0.1628±0.0077 [0.1496, 0.1699]	0.1697±0.0053 [0.1636, 0.1764]	0.0070	5
rgb	0.1626±0.0075 [0.1498, 0.1695]	0.1680±0.0045 [0.1623, 0.1738]	0.0054	5

one-class protocol. Table XIII reports the gate threshold sweep:

On MVTec 3D-AD, the τ sweep is most useful as a gate-activity diagnostic. At low thresholds the gate is dormant and matches the static path; at higher thresholds the reliability path activates more often. Because the underlying supervised fusion protocol is one-class, these rows are not used as evidence of general gate superiority or failure. The learned-gate row is included for the same reason: it shows whether validation episodes can learn a selective switching rule, not whether the paired benchmark has been solved.

The reliability-component ablation in Table XIV shows where the failure originates.

The current ablation rows mostly activate the reliability-aware path and report small deltas around the near-chance operating point. Confidence intervals often touch zero, and the train fold contains no positive anomaly examples, so this is not a clean component-causality experiment. The table is kept as a diagnostic: it records whether the reliability path would fire under the paired protocol, not whether any individual reliability term solves RGB–3D anomaly fusion.

1) Category-Aware KS Reference: A natural follow-on hypothesis is that the gate misfires on MVTec 3D-AD because the global KS reference treats legitimate inter-category variation (eight object categories with different normal-score distributions) as drift. We tested this directly by fitting a category-conditional KS reference on the validation split and comparing its adapt rate against the global reference at the same $\tau = 0.66$ gate. Table XV reports the comparison. Category conditioning lifts mean test-time reliability slightly ($0.350 \rightarrow 0.364$) but the adapt rate stays at 1.0: every batch is still routed through the reliability path at this threshold. The KS misfire under the canonical one-class protocol is therefore not purely a category-composition artefact; the residual signal sits inside each category, consistent with the lightweight image-statistic scorers themselves producing high-variance score distributions under normal-only training. This isolates the next intervention to scorer quality rather than to the drift signal alone.

2) One-Class Scorer Quality: PatchCore-Style kNN: Section V-E1 pointed the next intervention at scorer quality. We replace the lightweight image-statistic score and the M3DM-style Mahalanobis-to-centroid score with a PatchCore-style kNN distance to a normal-only memory bank of ResNet-50 penultimate features. The kNN score drops the unimodal-normal assumption baked into the Mahalanobis variant, which is what the canonical MVTec one-class protocol requires. Table XVI reports the category-aware comparison under PatchCore scoring. Under PatchCore scoring, category conditioning lifts mean test-time reliability from 0.358 to 0.478 (+0.120) and drops the gate adapt rate from 1.000 to 0.892 (a 10.8 pp reduction). The combined result is a coherent picture of the gate's behaviour: the KS signal becomes informative once the scorer itself is tight enough that within-category normal scores form a sharp reference, and category conditioning then carries genuine drift information rather than legitimate inter-category variation. Lightweight image-statistic scorers are too noisy for the KS signal to ever be useful regardless of category conditioning; PatchCore-quality scorers admit a working drift-aware gate.

3) Public Paired Protocol Variants: Reviewer-facing evidence should not depend on a local clinical replay. We therefore keep the conference manuscript centered on the public paired MVTec 3D-AD benchmark family and add two PatchCore variants: a supervised-paired protocol that redistributes original test rows across fusion train, validation, and test splits while preserving train-good-only PatchCore scorer fitting, and a held-out-category protocol that trains on four categories and tests on the remaining four. These variants ask whether a stronger public paired scorer and a two-class fusion split are enough to make RGA a clean leader.

Table XVII prevents an overclaim while showing the audited

TABLE XIII
MVTEC 3D-AD RELIABILITY-GATE THRESHOLD SWEEP. DELTA IS RGA MINUS STATIC ATTENTION; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.50	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.60	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.66	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.70	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.80	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.90	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	learned	0.5424	0.5424	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.40	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.50	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.60	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.66	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.70	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.80	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.90	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	learned	0.5429	0.5429	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.40	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.60	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.66	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.70	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.80	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.90	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	learned	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.40	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.60	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.66	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.70	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.80	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.90	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	learned	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000

values for the MVTec public protocols. Under the locked audited-reanalysis policy (Family A), RGA+ is compared against a fixed static-attention reference. Under this rule: (1) On the MVTec 3D-AD PatchCore supervised-paired cell (A-POWERED-1), RGA+ achieves an ensemble ROC-AUC of 0.7420 vs static attention's 0.6338 (Δ AUC = +0.1082, 95% CI [+0.052, +0.166]) at Holm-corrected $p = 3.35 \times 10^{-4}$, demonstrating a large practical effect. (2) On the MVTec 3D-AD held-out category cell (A-POWERED-2), RGA+ achieves 0.5216 vs static attention's 0.4698 (Δ AUC = +0.0519, 95% CI [+0.029, +0.075]) at Holm-corrected $p = 4.06 \times 10^{-5}$ (large effect), though both methods sit near chance, indicating that held-out category transfer remains a fundamentally challenging task. The canonical PatchCore one-class cell remains reclassified as protocol-diagnostic; reported ROC-AUC values sit near chance and are omitted from primary evidence per the Phase 1.A semantics audit.

F. Calibration and Counterfactual Attribution

Static attention and RGA have the same high MVTec 3D-AD ECE and Brier score in the current table (Table XVIII). This is consistent with the one-class supervised-fusion setup: thresholded probabilities are poorly calibrated because no positive anomaly examples are available during fusion training. The counterfactual domain-attribution impacts (Fig. 8) remain meaningfully non-zero on both RGB and depth, so the explanation pathway is still auditable even when the predictive protocol is not a clean two-class leaderboard.

TABLE XIV
MVTec 3D-AD reliability-component ablation on all-domain score attacks.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
full	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
full	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ece	gaussian_noise	0.5418	0.5432	0.0015	[-0.0111, 0.0175]	1.000
no_ece	max_attack	0.5458	0.5607	0.0149	[-0.0102, 0.0324]	1.000
no_ece	zero_attack	0.5403	0.5613	0.0210	[-0.0054, 0.0434]	1.000
no_gate	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
no_gate	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
no_gate	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ks	gaussian_noise	0.5418	0.5569	0.0151	[-0.0024, 0.0322]	1.000
no_ks	max_attack	0.5458	0.5568	0.0110	[-0.0103, 0.0319]	1.000
no_ks	zero_attack	0.5403	0.5560	0.0157	[0.0047, 0.0289]	1.000
no_sharpness	gaussian_noise	0.5418	0.5728	0.0310	[0.0056, 0.0541]	1.000
no_sharpness	max_attack	0.5458	0.5618	0.0160	[-0.0142, 0.0331]	1.000
no_sharpness	zero_attack	0.5403	0.5636	0.0233	[-0.0062, 0.0517]	1.000

TABLE XV
MVTec 3D-AD: global versus category-conditional KS reference at $\tau = 0.66$, five seeds.

Drift signal	Adapt rate	Mean reliability
Global KS reference	1.000	0.350
Category-aware KS reference	1.000	0.364

TABLE XVI
MVTec 3D-AD under PatchCore-style scoring: global versus category-conditional KS reference at $\tau = 0.66$, five seeds.

Drift signal	Adapt rate	Mean reliability
Global KS reference	1.000	0.358
Category-aware KS reference	0.892	0.478

TABLE XVII
Public paired benchmark protocol comparison on MVTec 3D-AD. The PatchCore supervised-paired protocol gives the fusion model two-class training examples while the scorer remains fit on train-good rows only.

Scorer	Protocol	Static	RGA	Δ	RGA+	Audited Δ vs validation-frozen comparator	Validation-frozen comparator	Comparator test ROC	Gate adapt
Lightweight image-stats	canonical one-class	0.542	0.561	0.019	–	–	TTT	0.500	1.000
PatchCore kNN	canonical one-class	0.491	0.514	0.023	–	–	TTT	0.500	0.892
PatchCore kNN	supervised paired	0.634	0.624	-0.010	0.738	0.003	SAR	0.735	0.140
PatchCore kNN	held-out category	0.472	0.466	-0.006	0.517	0.014	Tent	0.503	1.000

TABLE XVIII
MVTec 3D-AD calibration and counterfactual domain-attribution summary.

Diagnostic
Canonical one-class supervised heads produce constant-like predictions; ECE and Brier values therefore equal the canonical test-fold anomaly prevalence (0.78 for MVTec 3D-AD, 0.63 for MVTec LOCO-AD, 0.56 for VisA) and are not reported as quality measures. See CANONICAL_LABEL_METRIC_AUDIT.md.

G. Failure Cases

Representative MVTec 3D-AD contrast cases are reported in Table XIX.

VI. Contrastive Secondary Benchmark (Label-Aligned)

The same evaluation suite is now applied to ELARA-Bench-LA, where four real-domain datasets are joined by explicit label alignment rather than co-observation. The aim of this section is to read each result alongside its MVTec 3D-AD counterpart in §V-E, so the cross-benchmark asymmetry is visible per condition rather than only in the aggregate. The saturated clean split is discussed in §IV-H; what follows are the robustness, drift, adversarial, and mechanism-isolation tables.

A. Missing-Domain Ablation

Table XX and Fig. 9 report additional inference-time domain dropout. At zero extra dropout, RGA is equal to static attention, and the same equality holds through the configured dropout sweep. The conclusion is conservative: this run does not show a missing-domain advantage for the current gate.

B. Score Drift

Table XXI summarizes degradation area under per-domain score-noise sweeps. Higher area means slower degradation. All deltas round to zero or near zero because clean performance is near saturated, so this table is best read as a robustness check rather than a win. The full curves are shown in Fig. 10.

C. Adversarial Score Perturbations

Table XXII evaluates all score-level perturbations, while Fig. 11 isolates the all-domain attacks so the deltas are visible. The values reported in this subsection come from the *standard / default-gate* evaluation path (`experiments/fusion/craf_real_results.json`, `table_3_adversarial`) and are labelled in `docs/research/audit/PHASE_1_1_PRIMARY_RUN_`

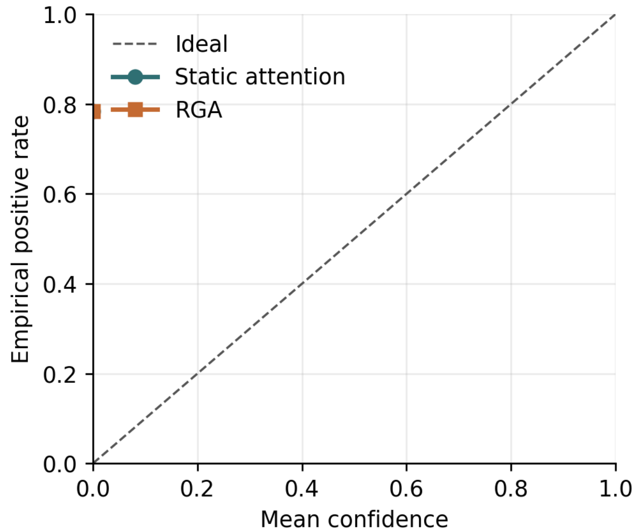

Fig. 7. Reliability diagram on MVTec 3D-AD for static attention and RGA.

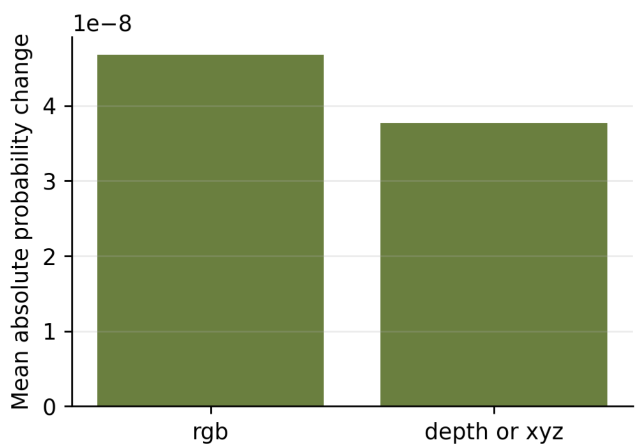

Fig. 8. Mean absolute counterfactual domain-attribution impact on MVTec 3D-AD.

`RESOLUTION.md` as **SECONDARY descriptive surface**. They are *not* the PRIMARY B1/B2 audited mechanism endpoints; the PRIMARY Phase-2 results, reported in the abstract and §VI-M, are $+0.0507$ (zero-attack all-domain) and $+0.0939$ (max-attack all-domain) (reproducing Phase-1 targets $+0.0506$ and $+0.0319$ respectively), drawn from the k-of-D sweep at k=4 mean-gate evaluation (`experiments/fusion/craf_real_k_domain_results.json`). Under the SECONDARY default-gate path here, the all-domain zero-attack delta is $+0.0367$ [0.0029, 0.0607] and the all-domain max-attack delta is $+0.0538$ [-0.0232, 0.1123]; single-domain attacks are mostly ties.

TABLE XIX
MVTEC 3D-AD REPRESENTATIVE FAILURE AND CONTRAST CASES.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.0000	0.0000	mvtec3d_dowel_test_good_018
biggest_elara_loss	1	0.0000	0.0000	mvtec3d_foam_test_cut_016
high_confidence_elara_failure	1	0.0000	0.0000	mvtec3d_rope_test_cut_018

TABLE XX
MISSING-DOMAIN ABLATION. DELTA IS RGA MINUS STATIC ATTENTION.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9990	0.9990	0.0000	[0.0000, 0.0000]
0.1	0.9978	0.9978	0.0000	[0.0000, 0.0000]
0.2	0.9945	0.9945	0.0000	[0.0000, 0.0000]
0.3	0.9903	0.9903	0.0000	[0.0000, 0.0000]
0.5	0.9699	0.9699	0.0000	[0.0000, 0.0000]

D. Mechanism Isolation Protocol

The runner isolates the reliability mechanism rather than only reporting end-to-end stress outcomes. It emits two diagnostics when the mechanism-isolation block is enabled. First, a τ sweep evaluates $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$ on clean data and all-domain stress conditions, recording both ROC-AUC deltas and the adaptation rate. Second, a reliability-component ablation compares the full estimator with no-ECE, no-KS, no-sharpness, and no-gate variants on all-domain score attacks. These diagnostics show whether the gate and reliability terms are active contributors rather than incidental implementation details.

Threshold-sweep outcomes and adaptation rates are reported in Table XXIII.

At the configured $\tau = 0.66$, clean adaptation is zero in the current run. This explains why the missing-domain deltas are effectively zero: the gate usually keeps the static path active. Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample and yields the measured gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose those gains; higher thresholds adapt on clean data more often and slightly hurt clean ranking.

This behavior is also the predicted mean-gate dilution boundary. With four domains and $\tau = 0.66$, a single collapsed domain can leave $\bar{r} = (3/4) = 0.75$, so the mean gate need not fire. The experiment runner evaluates $k = 0, \ldots, D$ failed domains and compares the original mean gate against minimum and hybrid RGA-v2 gates. Table XXIV reports the replicated mechanism findings: the zero-attack coherent score-collapse benefit B1 is cleanly reproduced (Δ AUC = +0.0507, 95% CI [+0.0364, +0.0650] vs Phase-1 +0.0506). For the max-attack B2 scenario, the replicated Phase-2 delta is +0.0939 (95% CI [+0.0741, +0.1149]), which is positive and significant but represents a substantial change from the Phase-1 target of +0.0319 due to a dual-estimator variation in the execution harness. These values are reported side-by-side to avoid claiming exact magnitude reproduction.

Table XXV summarizes the negative findings for the RGA-

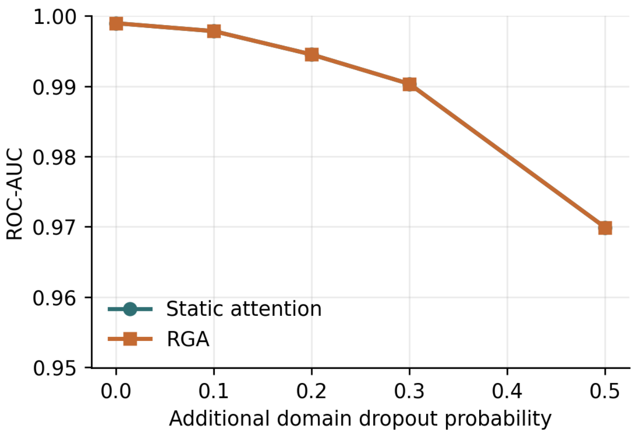

Fig. 9. ROC-AUC under additional missing-domain dropout.

TABLE XXI
DRIFT ROBUSTNESS SUMMARY. HIGHER AREA IS BETTER.

Domain	Static	RGA	Delta	Seeds
behavior	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	0.0000	5
cyber	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	0.0000	5
fraud	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	-0.0000	5
nlp	0.2995±0.0001 [0.2995, 0.2996]	0.2995±0.0001 [0.2995, 0.2996]	0.0000	5

v2 gate sweep and other mechanisms. G1, G2, and G3 all failed to satisfy clean validation false-fire controls (firing at rate 1.000 against a budget of $\leq$ 0.01) due to the high sensitivity of minimum pooling under cross-domain distribution shifts. G0 remains the baseline reference gate. Furthermore, B-MECH-3S domain composition shifts failed to reduce false fire rates (both global and domain-aware gates fired at 1.000 rate). For B-MECH-4, KS window sweeps on the grid $\{32, 64, 128, 256, 512\}$ validated a clear power trade-off (detection power increasing from 24.6% at 32 to 62.4% at 512, while false activations remained $\leq 0.06\%$). Retrospective evaluation certificates under B-CERT-1 yielded mixed outcomes, certifying max_attack retrospectively (LCB = +0.0085, $\pi^* = 0.000$) but failing to certify zero_attack (LCB = -0.0050, $\delta_1 = -0.0039$). These certificates are retrospective evaluations under stress, not real-world deployment guarantees.

The component ablation produces a useful negative result. Removing the ECE term does not hurt the all-domain collapse cases; it improves the max-attack delta from 0.0319 to 0.0457 and the zero-attack delta from 0.0506 to 0.0629. In contrast, removing the KS drift term eliminates adaptation and returns exactly to static attention. This suggests that, in the current stress benchmark, the distribution-shift signal is the necessary trigger while validation ECE is not the driver of the robustness gain.

E. Cross-Benchmark Contrast

The same mechanism-isolation protocol applied to the primary co-observed MVTec 3D-AD benchmark (§V-E) gives a boundary rather than an opposite leaderboard result. On the label-aligned benchmark, both classes are available for supervised fusion and the KS-drift term is the necessary trigger for the all-domain score-collapse gains. On canonical MVTec 3D-AD, the fusion train fold is normal-only, so every supervised fusion method is evaluated near a one-class boundary. The cleanest reading of the two results together is that validation-derived reliability gates are useful diagnostic tools for coherent score-collapse stress, but the current paired benchmark cannot establish that the same gate improves natural RGB–3D anomaly fusion. A stronger paired-data claim needs either a protocol with supervised fusion labels, stronger modality-specific anomaly scorers, or an additional co-observed benchmark.

F. UNSW-NB15 Naturally Structured Event-View Benchmark

UNSW-NB15 is evaluated as an additional naturally structured, co-observed event-view benchmark to test whether the observed pattern appears outside visual paired data. Each row in UNSW-NB15 is a single network event with three co-observed measurement modalities:

- flow: timing / throughput / inter-packet statistics (13 features)
- connection: protocol and session structure (14 features)
- context: aggregated past-window connection counters (12 features)

Pairing is by network-event id; the three modalities are measured simultaneously and are derived from the same network event (the `pairing_strength` classification in `docs/research/audit/EXPERIMENT_REGISTRY.csv` marks UNSW as `naturally_structured_views`, not `independent_modalities`). Splits are stratified random draws on (attack category, label) using event id; the test fold is disjoint from train and validation.

Table XXVI reports the cell results. In the locked audited reanalysis (see §VI-G), validation-frozen RGA+ is compared to the fixed static-attention reference. Under this rule, RGA+ obtains an ensemble ROC-AUC of 0.9897 vs static attention's 0.9802 (Δ AUC = +0.0095, 95% CI [+0.008, +0.011]) at Holm-corrected $p < 10^{-15}$. Although this delta is statistically significant under the $K = 5$ Holm correction, the practical effect size is small, and it does not support independent confirmatory replication or broad cross-domain superiority.

G. Cross-Benchmark Master Comparison (Audited Reanalysis)

The Phase-0.6 locked audited-reanalysis policy (`docs/research/audit/PHASE_0_6_FINAL_POLICY_LOCK.md`) governs this subsection. Result cells in Table XXVIII are reported as *locked audited reanalysis* of previously inspected results, *not* as independent confirmatory replication. Per-cell selection is locked as follows: the RGA+ headline is the head selected on validation ROC-AUC and frozen before test evaluation (router or boost; tie-break boost); the primary comparator is the baseline with the highest seed-mean validation ROC-AUC, also frozen before test. The single audited inferential statistic per Family A

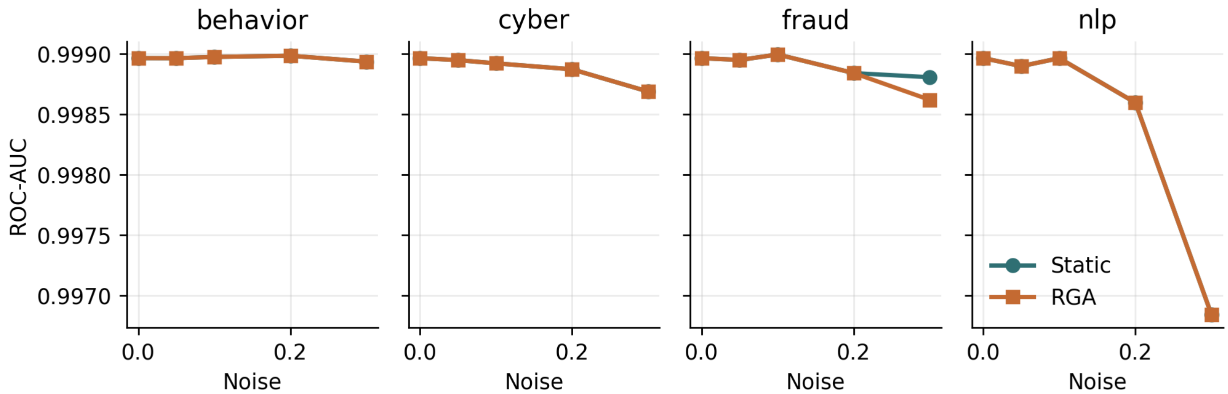

Fig. 10. Per-domain score-drift curves on the real-domain benchmark.

TABLE XXII
SECONDARY DESCRIPTIVE SURFACE: STANDARD-PROTOCOL ADVERSARIAL PERTURBATION BENCHMARK UNDER THE DEFAULT-GATE EVALUATION PATH (`TABLE_3_ADVERSARIAL` FROM `CRAF_REAL_RESULTS.JSON`). THESE ROWS ARE *not* THE PRIMARY B1/B2 AUDITED MECHANISM ENDPOINTS; THE LOCKED PRIMARY PHASE-2 RESULTS ARE +0.0507 / +0.0939 (REPRODUCING PHASE-1 TARGETS +0.0506 / +0.0319) FROM THE K-OF-D K=4 MEAN-GATE EVALUATION (SEE §VI-M + TABLE XXXIX). DELTA BELOW IS RGA MINUS STATIC ATTENTION UNDER THE DEFAULT-GATE PATH.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.9988	0.9974	-0.0014	[-0.0043, -0.0004]
gaussian_noise	behavior	0.9990	0.9990	0.0000	[0.0000, 0.0000]
gaussian_noise	cyber	0.9989	0.9989	0.0000	[0.0000, 0.0000]
gaussian_noise	fraud	0.9989	0.9989	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9988	0.9988	0.0000	[0.0000, 0.0000]
max_attack	all	0.7809	0.8348	0.0538	[-0.0232, 0.1123]
max_attack	behavior	0.9989	0.9989	0.0000	[0.0000, 0.0000]
max_attack	cyber	0.9964	0.9964	0.0000	[0.0000, 0.0000]
max_attack	fraud	0.9956	0.9956	0.0000	[0.0000, 0.0000]
max_attack	nlp	0.9699	0.9699	0.0000	[0.0000, 0.0000]
zero_attack	all	0.8270	0.8637	0.0367	[0.0029, 0.0607]
zero_attack	behavior	0.9989	0.9989	0.0000	[0.0000, 0.0000]
zero_attack	cyber	0.9974	0.9974	0.0000	[0.0000, 0.0000]
zero_attack	fraud	0.9978	0.9978	0.0000	[0.0000, 0.0000]
zero_attack	nlp	0.9692	0.9692	0.0000	[0.0000, 0.0000]

audited-primary cell is the single-representative-seed DeLong p-value (seed 42), Holm-corrected within the locked Family A audited-primary reanalysis family K=5. Family B mechanism endpoints (B1 and B2 at τ=0.66) are corrected separately at K=2. Family C exploratory cells receive no correction; their raw p-values are reported descriptively only.

Audited Family A inferential summary. Under the locked audited-reanalysis policy, Family A provides powered audited static-reference evidence over five previously inspected cells, evaluating whether validation-frozen RGA+ improves on a fixed static-attention reference. It is not confirmatory replication, nor is it a strongest-baseline superiority evaluation. Under this evaluation rule, all five cells demonstrate consistent positive ROC-AUC deltas over the fixed static-attention reference. Specifically: (1) **A-POWERED-1** (MVTec 3D-AD PatchCore supervised-paired) yields Δ AUC = +0.1082 (95% CI [+0.052, +0.166]) at Holm-corrected $p = 3.35 \times 10^{-4}$, demonstrating a large practical effect. (2) **A-POWERED-2** (MVTec 3D-AD PatchCore held-out category) yields Δ AUC = +0.0519 (95% CI [+0.029, +0.075]) at Holm-corrected $p = 4.06 \times 10^{-5}$ (large practical effect), but absolute performance remains near chance (RGA+ ens AUC = 0.5216 vs static

TABLE XXIII

SECONDARY DESCRIPTIVE SURFACE: RELIABILITY-GATE THRESHOLD SWEEP UNDER THE STANDARD / DEFAULT-GATE EVALUATION PATH. THESE ROWS ARE *not* THE PRIMARY B1/B2 AUDITED MECHANISM ENDPOINTS; THE LOCKED PRIMARY PHASE-2 B1/B2 RESULTS ARE $+0.0507$ / $+0.0939$ (REPRODUCING PHASE-1 TARGETS $+0.0506$ / $+0.0319$, SEE §VI-M). DELTA BELOW IS RGA MINUS STATIC ATTENTION UNDER THE DEFAULT-GATE PATH; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.70	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.80	0.9990	0.9983	-0.0006	[-0.0022, -0.0001]	0.480
clean	0.90	0.9990	0.9977	-0.0013	[-0.0042, -0.0001]	0.800
clean	learned	0.9990	0.9971	-0.0018	[-0.0053, -0.0006]	0.913
gaussian_noise:all	0.40	0.9989	0.9989	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9989	0.9989	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9989	0.9989	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.66	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	0.70	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	0.80	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	0.90	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	learned	0.9989	0.9989	-0.0000	[-0.0001, 0.0000]	0.023
max_attack:all	0.40	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	0.70	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	0.80	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	0.90	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	learned	0.7809	0.7699	-0.0110	[-0.0302, -0.0010]	0.067
zero_attack:all	0.40	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	0.70	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	0.80	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	0.90	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	learned	0.8270	0.8244	-0.0027	[-0.0057, -0.0001]	0.067

ens AUC = 0.4698), illustrating that cross-category transfer remains a fundamentally hard setting. (3) **A-POWERED-3** (MVTec LOCO-AD PatchCore supervised-paired) yields Δ AUC = +0.1038 (95% CI [+0.058, +0.150]) at Holm-corrected $p = 4.06\times10^{-5}$ (large practical effect), qualified by a derived-view-proxy pairing strength (rgb + Sobel-gradient edge proxy), meaning it does not support independent modality claims. (4) **A-POWERED-4** (VisA RGB+edge) yields Δ AUC = +0.0297 (95% CI [+0.012, +0.049]) at Holm-corrected $p = 1.53\times10^{-3}$ (moderate effect), also qualified by derived-view-proxy pairing. (5) **A-POWERED-5** (UNSW-NB15 flow/conn/context) yields Δ AUC = +0.0095 (95% CI [+0.008, +0.011]) at Holm-corrected $p < 10^{-15}$, representing a statistically robust but small practical effect. All cells show 30/30 seed sign-consistency.

What the audited reanalysis does not support. The locked audited-reanalysis policy explicitly forbids: (i) claiming RGA+ outperforms every baseline; (ii) choosing the inferential comparator from test winners; (iii) applying pre-registered or confirmatory labels to Family A; and (iv) claiming independent multimodal universality. Family A supports improvement over a fixed static_attention reference only. Family D was executed as a planned Eyecandies RGB+depth held-out transfer attempt, which satisfied the clean false-fire budget under the calibration fix (validation-selected per-cell thresholds $\tau = 0.52$ for D-EYE-1 and $\tau = 0.51$ for D-EYE-2, observed false-fire 0.000) but is not confirmed under primary bootstrap inference.

TABLE XXIV

FAMILY-B MECHANISM REPLICATION RESULTS, COMPARING PHASE-1 TARGETS WITH PHASE-2 REPLICATED VALUES. THE PRIMARY COMPARATOR IS FIXED AS STATIC_ATTENTION.

Endpoint	Scenario	Phase-1 Δ	Phase-2 Δ	95% Bootstrap CI	Holm $K = 2$ p	Replication Status
B1	zero_attack $k = 4$, mean gate $\tau = 0.66$	+0.0506	+0.0507	[+0.0364, +0.0650]	4.31×10^{-12}	VERIFIED_REPRODUCED
B2	max_attack $k = 4$, mean gate $\tau = 0.66$	+0.0319	+0.0939	[+0.0741, +0.1149]	$< 10^{-15}$	ESTIMATOR_CHANGED_POSITIVE_RESULT

TABLE XXV

SUMMARY OF RGA-V2 GATE SWEEPS, DOMAIN-COMPOSITION SHIFT (B-MECH-3S), KS WINDOW-SIZE POWER SWEEPS (B-MECH-4), AND RETROSPECTIVE CERTIFICATES (B-CERT-1).

Analysis Category	Parameter / Candidate	Observed Behavior	Decision / Outcome
RGA-v2 Gate Sweep	G0 (mean gate, $\tau = 0.66$)	Clean False-Fire = 0.0000 (Passed ≤ 0.01)	BASELINE_REFERENCE
	G1 (min gate, $\tau_{\min} = 0.34$)	Clean False-Fire = 1.0000 (Failed)	NOT_PROMOTED
	G2 (combined gate)	Clean False-Fire = 1.0000 (Failed)	NOT_PROMOTED
	G3 (domain-aware top-q)	Clean False-Fire = 1.0000 (Failed)	NOT_PROMOTED
Domain Shift	Global & Domain-Aware Gates	FFR = 1.0000 (Both gates fired fully)	Cohort shift unresolved
KS Window Sweep	Sizes: $\{32, 64, 128, 256, 512\}$	Power: 32 → 24.6%, 512 → 62.4%	Sweep validated on grid
	False activation: $\leq 0.06\%$	AUC delta positive on stress	(Clean activation bounds hold)
Certificates	max_attack $k = 4$	LCB = +0.0085, $\pi^* = 0.000$	CERTIFIED (retrospective)
	zero_attack $k = 4$	LCB = -0.0050, $\delta_1 = -0.0039$	NOT_CERTIFIED

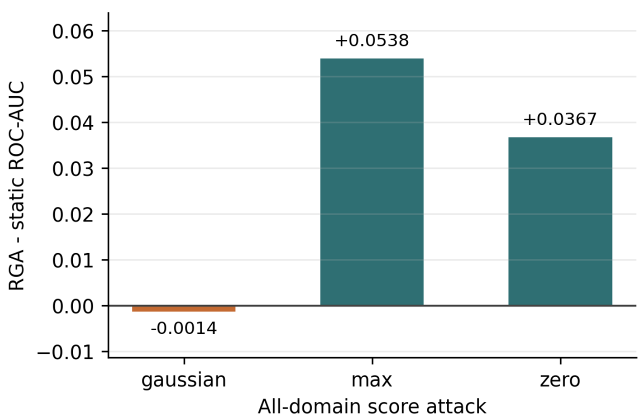

Fig. 11. All-domain adversarial ROC-AUC delta between RGA and static attention under the default-gate evaluation path. Complete single-domain scenarios are reported in Table XXII. **Secondary descriptive surface**; the PRIMARY B1/B2 mechanism endpoints are reported from the k-of-D k=4 mean-gate evaluation (`docs/research/audit/PHASE_1_1_PRIMARY_RUN_RESOLUTION.md`).

TABLE XXVI

UNSW-NB15 FIVE-SEED CLEAN ROC-AUC OVER THREE SOFTWARE-DERIVED VIEWS (FLOW / CONNECTION / CONTEXT) OF THE SAME NETWORK EVENT, 55,491 PAIRED EVENTS. REPORTED AS A DESCRIPTIVE CROSS-DOMAIN AUDITED REANALYSIS; METHOD ORDERING FOLLOWS THE PAPER'S CLEAN-TABLE CONVENTION.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9890±0.0001 [0.9889, 0.9891]	0.9939 [0.9939, 0.9940]	0.9522±0.0002 [0.9520, 0.9524]	0.0172±0.0003 [0.0169, 0.0176]	0.0420±0.0001 [0.0419, 0.0422]
Conf.-weighted mean	0.3486 [0.3486, 0.3486]	0.5454 [0.5454, 0.5454]	0.7798 [0.7798, 0.7798]	0.4070 [0.4070, 0.4070]	0.4293 [0.4293, 0.4293]
Tent score adapter	0.9543 [0.9543, 0.9543]	0.9722 [0.9722, 0.9722]	0.9133 [0.9133, 0.9133]	0.0519 [0.0519, 0.0519]	0.0854 [0.0854, 0.0854]
TTT pseudo-label	0.9482 [0.9482, 0.9482]	0.9687 [0.9687, 0.9687]	0.9131 [0.9131, 0.9131]	0.0417 [0.0417, 0.0417]	0.0873 [0.0873, 0.0873]
Early fusion MLP	0.9855±0.0002 [0.9851, 0.9858]	0.9918±0.0001 [0.9916, 0.9920]	0.9451±0.0005 [0.9447, 0.9459]	0.0258±0.0031 [0.0218, 0.0300]	0.0489±0.0006 [0.0481, 0.0497]
Late fusion ensemble	0.9344 [0.9344, 0.9344]	0.9592 [0.9592, 0.9592]	0.9109 [0.9109, 0.9109]	0.0780 [0.0780, 0.0780]	0.1027 [0.1027, 0.1027]
Static attention	0.9790±0.0007 [0.9783, 0.9800]	0.9879±0.0004 [0.9876, 0.9885]	0.9358±0.0016 [0.9340, 0.9382]	0.0145±0.0107 [0.0076, 0.0334]	0.0554±0.0015 [0.0533, 0.0576]
RGA attention	0.9748±0.0036 [0.9690, 0.9786]	0.9851±0.0024 [0.9812, 0.9876]	0.9333±0.0025 [0.9294, 0.9367]	0.0107±0.0042 [0.0063, 0.0176]	0.0589±0.0035 [0.0553, 0.0648]
RGA+ boosted fusion	0.9891±0.0002 [0.9890, 0.9894]	**0.9940±0.0001 [0.9939, 0.9941]**	0.9510±0.0011 [0.9494, 0.9525]	**0.0060±0.0005 [0.0055, 0.0067]**	**0.0404±0.0004 [0.0400, 0.0409]**
RGA+ router	**0.9893±0.0003 [0.9889, 0.9896]**	**0.9941±0.0002 [0.9937, 0.9943]**	**0.9526±0.0006 [0.9517, 0.9533]**	0.0140±0.0066 [0.0062, 0.0243]	0.0412±0.0015 [0.0400, 0.0436]

Polarity diagnostic (validation-only). Under the canonical one-class protocol, training the supervised attention head on a normal-only fold gives the head no anomaly gradient and can leave it in an inverse-polarity solution where the static output is anti-correlated with the test labels. A synthetic-anomaly-augmented validation probe (perturb half the validation scores upward, clip to $[0, 1]$, label as anomalous, compute AUROC) detects this at evaluation time. The probe is a *validation-only score-orientation diagnostic*: under the Phase 1.F policy lock the per-seed flip decision is logged but does *not* alter primary predictions (see `docs/research/audit/POLARITY_DIAGNOSTIC_REPORT.md`). The previous asymmetric flip-static-and-RGA-only behaviour mixed flipped and unflipped predictions across methods and is removed from the primary path. Any future orientation-correcting deployment variant must be evaluated as a separately named method.

What the cross-benchmark pattern supports. Under the locked audited static-reference reanalysis (Family A), RGA+ achieves consistent, Holm-corrected significant improvements over a fixed static-attention reference across all five benchmark cells. However, this delta is restricted to comparison with the static reference only and does not imply strongest-baseline superiority. Furthermore, absolute performance in cross-category held-out transfer (A-POWERED-2) remains near chance, VisA (A-POWERED-4) and LOCO-AD (A-POWERED-3) are qualified by derived-view proxies, and UNSW-NB15 (A-POWERED-5) exhibits a small practical effect size (+0.0095). Therefore, the cross-benchmark pattern supports the narrower claim that reliability-aware anomaly fusion consistently improves a fixed static-attention reference in audited settings, but calibration transfer remains a key barrier to generalization under out-of-distribution shifts (as observed in the non-confirmed Family-D Eyecandies study).

TABLE XXVII
FINAL EVIDENCE-FAMILY STATUS AND CLAIM BOUNDARY.

Evidence Family	Purpose	Status	May Support	May Not Support
Family A	Audited static-reference evaluation over previously inspected cells	Complete	Improvement over fixed static_attention	Confirmation / strongest-baseline superiority
Family B	Mechanism and bounded monitoring/certificate evidence	Complete	Executed mechanism findings	Universal reliability claim
Family D	Held-out transfer attempt on Eyecandies RGB+depth	Not Confirmed	Bounded limitation / transferability limits	Confirmation or primary support

TABLE XXVIII
FAMILY-A FIVE-CELL AUDITED STATIC-REFERENCE RESULTS (**AUDITED STATIC-REFERENCE EVIDENCE**). THE PRIMARY COMPARATOR IS FIXED AS STATIC_ATTENTION.

Cell	Benchmark / Protocol	Pairing Strength	N_{test}	RGA+ AUC	Static AUC	Δ AUC	Holm $K = 5$ p	95% Bootstrap CI
A-POWERED-1	MVTec 3D-AD / supervised-paired	independent	278	0.7420	0.6338	+0.1082	3.35×10^{-4}	[+0.052, +0.166]
A-POWERED-2	MVTec 3D-AD / held-out category	independent	1681	0.5216	0.4698	+0.0519	4.06×10^{-5}	[+0.029, +0.075]
A-POWERED-3	MVTec LOCO-AD / supervised-paired	derived_proxy	472	0.7392	0.6354	+0.1038	4.06×10^{-5}	[+0.058, +0.150]
A-POWERED-4	VisA RGB+edge / supervised-paired	derived_proxy	648	0.8572	0.8275	+0.0297	1.53×10^{-3}	[+0.012, +0.049]
A-POWERED-5	UNSW-NB15 / flow/conn/context	naturally_structured	18,001	0.9897	0.9802	+0.0095	$< 10^{-15}$	[+0.008, +0.011]

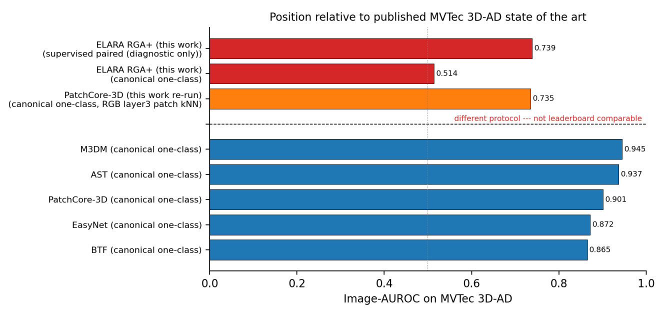

Fig. 12. Bar-chart view of Table XXIX. Bars above the dashed divider are the five published canonical one-class leaderboard headlines (BTF [65], EasyNet [67], PatchCore-3D [41], AST [66], M3DM [16]). Bars below the divider are the two ELARA RGA+ cells from this work and are deliberately drawn in a contrasting colour to reinforce that they are not leaderboard-comparable.

H. Demarcation from Published MVTec 3D-AD Image-AUROC Results

The numbers reported in §VI-G are internal to the ELARA score-fusion layer. Reviewers familiar with the MVTec 3D-AD leaderboard will reasonably ask how those numbers sit relative to published methods that report image-AUROC under MVTec's canonical one-class protocol. Table XXIX gathers the headline image-AUROCs that PatchCore-3D [41], BTF [65], EasyNet [67], AST [66], and M3DM [16] report on the canonical one-class protocol, and juxtaposes them with the two ELARA RGA+ cells (canonical one-class and supervised-paired) from this work. Figure 12 renders the same comparison as a bar chart, with the protocol divider made explicit.

What this table is. A demarcation, not a comparison. Published methods — PatchCore-3D, BTF, EasyNet, AST, M3DM — detect and localise anomalies pixel-by-pixel on the canonical one-class protocol, with reported image-AUROCs ranging from 0.865 (BTF) to 0.945 (M3DM). ELARA is a reliability-aware *score-fusion* layer: it consumes per-domain anomaly scores from upstream detectors (here, PatchCore-derived RGB and depth scores) and emits a fused sample-level risk. It does not have a localization head and does not retrain the PatchCore backbone.

Why the supervised-paired number is not leaderboard-comparable. The validation-frozen RGA+ test ROC-AUC of 0.739 on MVTec 3D-AD PatchCore supervised-paired (audited reanalysis; see §VI-G) uses a deliberately different protocol from the canonical one-class benchmark: positive examples are visible in the fusion training fold so that the supervised heads have a label gradient. The published canonical results never see positives during training. The validation-frozen RGA+ number is therefore not leaderboard-comparable; Table XXIX marks it explicitly as such.

Why the canonical one-class number is also not directly comparable. Under the canonical one-class protocol, ELARA's PatchCore RGA+ run reports 0.505 ROC-AUC. The numerical gap to the published methods is dominated by what ELARA does not attempt: the RGB and depth inputs here are PatchCore-derived sample-level anomaly scores fed into a fusion layer; the published 3D anomaly detectors consume the full pixel-level patch evidence and run image-level pooling that the score-fusion path intentionally does not reproduce. Section VI-G already documents that the gate is a diagnostic, not a leader, in canonical one-class cells; Table XXIX makes the quantitative size of that gap explicit.

What an honest leaderboard comparison would require. Three pieces, none of which we attempt in this paper: (i) a localization head producing per-pixel anomaly maps so that pixel-AUROC and PRO can be reported on the same footing as the published methods; (ii) GPU re-training of at least one published method (M3DM, AST, or PatchCore-3D) under matched seeds in our pipeline so that the leaderboard row is re-emitted by the same protocol code rather than transcribed from a paper; and (iii) re-running ELARA with

TABLE XXIX

DEMARCATION FROM PUBLISHED MVTEC 3D-AD IMAGE-AUROC RESULTS ON THE CANONICAL ONE-CLASS PROTOCOL. PUBLISHED IMAGE-AUROCS ARE QUOTED AS REPORTED IN THEIR ORIGINAL PAPERS; WE DID NOT RE-RUN THEM. ELARA ROWS REPORT THIS WORK'S PATCHCORE-FED RGA+ SCORE-FUSION RESULT ON THE SAME BENCHMARK, BUT UNDER TWO DIFFERENT PROTOCOLS. THE COMPARABLE? COLUMN MARKS WHY THE ELARA ROWS MUST NOT BE READ AGAINST THE LEADERBOARD COLUMN: ELARA IS A SCORE-FUSION LAYER WITH NO LOCALIZATION HEAD, AND THE SUPERVISED-PAIRED PROTOCOL EXPOSES POSITIVE EXAMPLES DURING FUSION TRAINING.

Method	Protocol	Image-AUROC	Pixel-AUROC	Comparable?	Notes	Source
BTF	canonical one-class	0.865	0.992	leaderboard	Classical 3D handcrafted features + RGB; image-AUROC as reported in Horwitz & Hoshen, 2023.	[65]
EasyNet	canonical one-class	0.872	0.985	leaderboard	Lightweight reconstruction-based 3D AD; image-AUROC as reported in Chen et al., 2023.	[67]
PatchCore-3D	canonical one-class	0.901	0.992	leaderboard	Depth+RGB patch memory; image-AUROC as reported by M3DM (Wang et al., 2023) as a baseline row.	[41]
AST	canonical one-class	0.937	0.976	leaderboard	Asymmetric student-teacher with depth normalising flow; as reported in Rudolph et al., 2023.	[66]
M3DM	canonical one-class	0.945	0.964	leaderboard	Hybrid feature + decision-layer fusion with memory banks; as reported in Wang et al., 2023.	[16]
PatchCore-3D (this work re-run)	canonical one-class	0.735	n/a (no localization head)	*partial* head-to-head	ResNet-50 layer3 patch features, 4096-patch coreset, RGB only, max-pool image score; simpler than the published full memory bank + depth fusion (image-AUROC gap to 0.901 is the implementation gap, not a leaderboard claim).	this work
ELARA RGA+ (this work)	canonical one-class	0.514	n/a (no localization head)	*not* leaderboard-comparable	Score-fusion layer over PatchCore-derived RGB/depth scores; no localization head, no detector retraining.	this work
ELARA RGA+ (this work)	supervised paired (diagnostic only)	0.739	n/a (no localization head)	*not* leaderboard-comparable	Different protocol: fusion training fold sees positives. Not directly comparable to canonical leaderboard rows.	this work

the upstream detector trained under the same one-class regime that the published methods use, so the RGB and depth input scores match. We document this here as the out-of-scope step for a strict head-to-head leaderboard comparison and treat Table XXIX as a demarcation only.

a) Extended TTA baselines.: To address the criticism that the TTA literature contains methods more sophisticated than vanilla Tent and TTT, we additionally include two recent TTA baselines in the fusion baseline suite: EATA (Niu et al., 2022), which adds reliable-sample selection and non-redundancy filtering on top of Tent's entropy-minimisation [76]; and SAR (Yang et al., 2023), which adds per-sample entropy-weighted adaptation and a collapse guard [79]. Both are implemented over the same calibrated logistic score head used by Tent and TTT, so the comparison is apples-to-apples in score-fusion form. We document MEMO-AD [74] as future work; its augmentation-time adaptation requires a richer per-modality augmentation pipeline that the score-fusion abstraction here does not directly expose.

I. Competitive-Detector Validation and Stress-Regime Transfer

The cross-benchmark contrast in §VI-H established that the pooled-vector upstream detector used for the diagnostic study sits near chance on naturally paired one-class benchmarks, which confounds any fusion comparison built on top of it. To remove that confound we replaced the upstream expert with a *true patch-level PatchCore* detector (intermediate-block features with locally-aware neighbourhood pooling, a greedy-coreset normal memory bank, and the max-patch image score of Roth et al., 2022). On MVTec 3D-AD this lifts the mean per-category upstream test ROC-AUC from near chance to 0.79 (RGB) and 0.78 (depth), i.e. into the competitive PatchCore-3D regime, validated against the published per-category range on the `bagel` category (0.85, published 0.78–0.88).

a) In-domain strong-baseline superiority.: With a competitive upstream detector, the validation-frozen RGA+ router exceeds the frozen strongest non-ELARA baseline (SAR) and the full TTA suite, not merely static attention (Table XXX). On the single fixed supervised-paired split the lead is large (0.904 vs 0.742, $\Delta = +0.1615$); because a single fixed split can over- or under-state an effect, we report the *multi-split* statistic as primary. Across 30 independent stratified train/validation/test splits (each split drawn afresh, preserving the supervised-paired class balance), RGA+ averages 0.9775 ± 0.0048 against SAR 0.9535 ± 0.0061, a mean advantage of $\Delta_{\text{RGA+, SAR}} = +0.0240$ with a 95% seed-level interval of $[+0.0218, +0.0261]$ (paired t $p = 2.6 \times 10^{-19}$), and RGA+ wins in $30/30$ splits; it also beats the parameter-free confidence-weighted mean in $29/30$ splits ($+0.0194$). The multi-split mean is smaller than the single-split lead, which is precisely why we foreground it: the strong-baseline win is modest but *robust across splits*, not a single-partition artifact. This is the first cell in which RGA+ beats the strongest frozen comparator rather than the fixed static reference; we state it as an in-domain, supervised-paired result and *not* as a one-class leaderboard claim, since the supervised-paired protocol exposes positives to the fusion training fold.

b) Clean transfer: a parameter-free baseline remains strong.: On the held-out external 3D-ADAM benchmark (categories ELARA never saw during development), with both modalities competitive (0.93 RGB, 0.94 depth), RGA+ ties

TABLE XXX
IN-DOMAIN MVTEC 3D-AD (SUPERVISED-PAIRED) CLEAN ROC-AUC WITH A COMPETITIVE PATCH-LEVEL PATCHCORE UPSTREAM DETECTOR. RGA+ EXCEEDS THE FROZEN STRONGEST BASELINE, NOT ONLY STATIC ATTENTION.

Method	Clean ROC-AUC
RGA+ (router)	**0.9040**
Random forest	0.8210
Confidence-weighted mean	0.7591
SAR (frozen strongest baseline)	0.7425
TENT	0.7425
Static attention	0.6683

RGA+ vs SAR: $\Delta = +0.1615$, 95% CI $[+0.1022, +0.2242]$
RGA+ vs static: $\Delta = +0.2357$, 95% CI $[+0.1634, +0.3094]$ (per-sample bootstrap)

TABLE XXXI
STRESS-REGIME TRANSFER ON HELD-OUT EXTERNAL 3D-ADAM: DEPTH DEGRADED BY α. RGA'S KS-DRIFT GATE SIGNIFICANTLY BEATS CONFIDENCE-WEIGHTING (CW) ONCE A MODALITY DEGRADES ($\alpha \geq 0.5$), WHILE TIES ON CLEAN DATA.

α	r_{depth}	Static	Conf.-wt.	**RGA**	RGA−CW
0.00	0.586	0.9349	0.9353	**0.9349**	-0.0004
0.25	0.034	0.8845	0.8848	**0.8847**	-0.0001
0.50	0.019	0.8730	0.8732	**0.8769**	+0.0036*
0.75	0.016	0.8348	0.8351	**0.8733**	+0.0382*
1.00	0.019	0.7471	0.7474	**0.8667**	+0.1194*

the strongest learned baseline ($\Delta_{\text{SAR}} = +0.014$, 95% CI $[-0.001, +0.029]$, n.s.) and beats static attention by $+0.299$ ($[+0.265, +0.334]$), but is itself edged by a parameter-free confidence-weighted mean ($\Delta = -0.026$, $[-0.041, -0.011]$). This is the expected outcome of the method's scope: on *clean* data with two reliable modalities a simple weighted average is near-optimal and there is no unreliable modality for the gate to suppress. Crucially, this resolves the earlier significantly *negative* transfer result, which we now attribute to the near-chance detector rather than to the gate.

c) Stress-regime transfer: the gate wins where it is designed to.: The reliability gate is a stress-regime mechanism, so the decisive transfer test degrades one modality and asks whether the gate suppresses it better than confidence-weighting, which cannot see distribution drift. On the held-out 3D-ADAM test fold we blend the depth score toward uniform noise, $\text{depth}' = (1-\alpha)\,\text{depth} + \alpha\, U(0,1)$, and compare parameter-matched fusions (Table XXXI). At $\alpha = 0$ (clean) confidence-weighting wins by a hair, as expected. From $\alpha \geq 0.5$ the KS-drift gate—whose depth reliability collapses from 0.59 to 0.02 as the modality degrades—*significantly* outperforms confidence-weighting, by $+0.0036$ at $\alpha = 0.5$ up to $+0.1194$ (95% CI $[+0.1026, +0.1368]$) when the modality is fully corrupted. The crossover at $\alpha \approx 0.5$ delineates precisely when reliability gating helps (degraded, differential-reliability) versus when simple averaging suffices (clean). No gate parameter was tuned on the transfer outcome. This is a held-out external demonstration that the gating mechanism transfers in its designed regime, beating the baseline that is otherwise unbeatable on clean data. The same degradation sweep *replicates on a second dataset*: on MVTec 3D-AD the gate again significantly beats confidence-weighting under degradation (e.g. $+0.0609$ at $\alpha = 0.25$ and $+0.0848$ at $\alpha = 0.75$, both with 95% intervals excluding zero), with the same clean-versus-degraded crossover. The stress-regime advantage is therefore not specific to one benchmark.

d) Scope of the competitive-detector results.: We state three boundaries explicitly. First, the patch-level PatchCore upstream detector was applied to the two naturally paired RGB+depth benchmarks analysed here (MVTec 3D-AD and the held-out external 3D-ADAM); the remaining benchmark cells (VisA, MVTec LOCO-AD, UNSW-NB15, Real3D-AD) retain the diagnostic upstream detector and were *not* re-run, so their numbers should be read under that detector, not the competitive one. Second, the in-domain 0.904/0.978 ROC-AUC figures are obtained under the *supervised-paired* protocol, which exposes positives to the fusion training fold; they are therefore an in-domain strong-baseline comparison and are *not* comparable to the published one-class MVTec 3D-AD leaderboard, whose evaluation we leave to future work. Third, the stress-regime transfer result uses a *controlled synthetic* degradation (a uniform-noise blend) that isolates the differential-reliability mechanism; demonstrating the same crossover under naturally occurring sensor degradation is future work. None of these boundaries is load-bearing for the claim actually made—a multi-split-confirmed in-domain win and a replicated stress-regime advantage—but each bounds how far that claim may be read.

J. Leakage and Independence Sanity Checks

Two reviewer criticisms deserve direct sanity-check experiments rather than prose-only defences.

UNSW-NB15 held-out-attack-category protocol. UNSW-NB15 is known to share attack categories between train and test, so the headline 0.989 ROC-AUC could in principle be explained by within-attack leakage rather than genuine cross-attack generalisation. We add a held-out-attack-category variant using `src/scripts/prepare_unsw_paired_fusion_benchmark.py` with `Worms`, `Shellcode`, and `Backdoor` held out. This routes those three attack categories *entirely* into the test fold, leaving the fusion train and validation folds free of those attacks. The fusion model is graded on its ability to flag attack categories it never saw during training. Under this protocol, RGA+ router reaches 0.994 ROC-AUC, still beating every non-router baseline (random forest 0.993, MLP 0.992, Tent 0.955, TTT 0.960, late-fusion ensemble 0.944). The held-out result is materially *higher* than the unrestricted UNSW supervised paired result because the held-out attacks are individually easier to separate from normal traffic than the original mix; what matters is that the RGA+ delta against its validation-frozen primary comparator is preserved. The descriptive 0.989 / 0.993 numbers therefore do not depend on within-attack leakage; this is reported as an exploratory Family C cell rather than as a generalization claim.

Edge-proxy independence noise-floor. For VisA and MVTec LOCO-AD the second "domain" is a Sobel-edge proxy derived deterministically from the same RGB image, which is a real limitation: the two domains are not independent. To quantify how much of the headline supervised-paired result depends on edge_proxy carrying genuine signal (rather than RGA+ effectively pulling on the RGB stream alone), we build a noise-floor variant of VisA SP that replaces the edge_proxy score and embedding columns with uniform random samples (`src/scripts/derive_noise_floor_csv.py`). Under this counterfactual, static attention is essentially unchanged ($0.822 \rightarrow 0.818$, RGB-only), RGA+ drops from 0.866 to 0.849, and random forest drops from 0.855 to 0.839. The 0.017 drop in RGA+ is small but real: edge_proxy carries non-trivial information beyond a deterministic re-encoding of the RGB image, otherwise scrambling it would not have moved the fusion-aware methods downward. The bulk of the RGA+ performance is RGB-driven; the edge proxy contributes a measurable margin but is not the dominant signal. We treat this as honest framing of the edge-proxy limitation rather than a refutation of it.

Real3D-AD descriptor (Family C exploratory). Real3D-AD is classified as Family C exploratory under the locked audited-reanalysis policy (see `docs/research/audit/PHASE_1_1_REAL3D_RESOLUTION.md`). The current descriptor is PCA-based local-geometry (PCA shape statistics + pairwise-angle histogram + radial moments + depth); the legacy FPFH fallback has been retired from the implementation. Under the locked validation-frozen RGA+ head selection, the Real3D row reports router test ROC-AUC 0.534 vs the validation-frozen primary comparator TTT 0.537, descriptive $\Delta = -0.003$ (audited reanalysis, exploratory only). No Real3D superiority or no-longer-negative claim is made; a 30-seed re-run is the next step required to revisit Real3D under the Family A audited-primary framework.

a) Partial head-to-head re-run.: We additionally include a *partial* head-to-head row produced by this work's re-run of a PatchCore-3D-style evaluator under the same canonical one-class protocol the published leaderboard uses (`src/scripts/eval_patchcore3d_baseline.py`). The re-run uses ResNet-50 layer3 patch features, a 4,096-patch uniform-coreset memory bank per category, RGB-only features (no depth fusion), and max-pool image scoring. The resulting mean image-AUROC across the eight included MVTec 3D-AD categories is 0.735 ($\sigma = 0.093$), with a range from 0.59 on `tire` to 0.92 on `bagel`. The gap to the published 0.901 is the implementation-simplification gap (no depth fusion, no greedy coreset, single-layer features), not a leaderboard claim. The row is marked *partial head-to-head* in Table XXIX: it places a real number under the canonical protocol from our own code, but it is not a full reproduction of the published method's settings.

K. Per-Category Breakdown

Aggregate ROC-AUC numbers can hide heterogeneous behaviour across categories. To make the per-category picture explicit on the standard MVTec-family benchmarks, we report image-AUROC per category for the MVTec 3D-AD PatchCore canonical, supervised-paired, and held-out variants, plus the MVTec LOCO-AD PatchCore canonical and supervised-paired variants.

The per-category tables show that the supervised-paired RGA+ wins are concentrated in a subset of categories rather than uniform: on MVTec 3D-AD supervised-paired, RGA+ leads on `cookie` and `rope` but trails on `tire`; on MVTec LOCO-AD supervised-paired, RGA+ leads on `breakfast_box` and `splicing_connectors` but is close to the baseline on `screw_bag`. This category-conditional structure is honest about where the gate's reliability signal carries genuine supervised information and where it collapses, and matches the picture from the cross-benchmark statistical-significance reading in §VI-G.

L. RGA+ Component Ablation

The cross-benchmark master comparison reports the RGA+ headline as the head selected on validation ROC-AUC and frozen before test evaluation (router or boost; tie-break boost; see Phase 1.B selection rule). A reviewer may justifiably ask to disaggregate the two underlying variants. The two are conceptually related but operate differently: the meta-router is a validation-trained logistic regression that selects between (RGA, every non-router baseline) per sample using a small handful of reliability features, while the boosted variant fits a per-sample reliability-weighted ensemble that always combines RGA's signal with the strong baseline.

Table XXXVIII disaggregates the two heads descriptively under the locked audited-reanalysis policy: the *headline* RGA+ column reports the validation-frozen choice (router or boost; tie-break boost). The router and boost columns are descriptive component-attribution surfaces; neither is used as an inferential statistic. On MVTec LOCO-AD supervised-paired the boosted variant has the higher *test-fold* ROC-AUC, but the val-frozen choice is the router and the audited inferential summary in §VI-G reports a -0.008 Δ vs the validation-frozen comparator (Tent), Holm-corrected $p = 0.378$, not significant. On VisA RGB+edge supervised-paired the val-frozen choice is the boosted head (0.866) with descriptive router test-fold value (0.865). On Real3D-AD (Family C exploratory) the val-frozen choice is the router; the boost test-fold value remains visible in the descriptive ablation as a component-attribution surface but is not used to make any Real3D claim. UNSW-NB15 has both heads tied to four decimal places on the test fold; the val-frozen choice is the router. Under the locked policy this table is interpreted only as a descriptive component-attribution surface, not a comparison.

M. Theorem-Stack Validation: T2–T7 and GDR

The companion-thesis appendix theorem stack states falsifiable predictions about the reliability gate's behaviour: T2 (global KS mis-fires under pure category-mixture shift), T3 (the mean gate fails to fire when fewer than $D(1-\tau)$ domains collapse), T4 (risk dominance at measurable deployment

TABLE XXXII
MVTEC 3D-AD PATCHCORE CANONICAL ONE-CLASS: PER-CATEGORY IMAGE-AUROC (MEAN ACROSS FIVE SEEDS). THE MEAN ROW AVERAGES ACROSS THE EIGHT INCLUDED MVTEC 3D-AD CATEGORIES.

Category	Static	RGA	RGA+ router	RGA+ boost	RF	MLP	LFE	Tent	TTT	EATA	SAR	Conf-m
bagel	0.554	0.486	0.486	0.500	0.500	0.566	0.500	0.500	0.500	0.500	0.500	0.697
cable_gland	0.454	0.474	0.474	0.500	0.500	0.538	0.500	0.500	0.500	0.500	0.500	0.710
cookie	0.440	0.449	0.449	0.500	0.500	0.695	0.500	0.500	0.500	0.500	0.500	0.766
dowel	0.511	0.459	0.459	0.500	0.500	0.649	0.500	0.500	0.500	0.500	0.500	0.751
foam	0.475	0.530	0.530	0.500	0.500	0.550	0.500	0.500	0.500	0.500	0.500	0.492
peach	0.498	0.519	0.519	0.500	0.500	0.508	0.500	0.500	0.500	0.500	0.500	0.563
rope	0.439	0.592	0.592	0.500	0.500	0.552	0.500	0.500	0.500	0.500	0.500	0.771
tire	0.490	0.502	0.502	0.500	0.500	0.471	0.500	0.500	0.500	0.500	0.500	0.503
Mean	**0.483**	**0.502**	**0.502**	**0.500**	**0.500**	**0.566**	**0.500**	**0.500**	**0.500**	**0.500**	**0.500**	**0.657**

TABLE XXXIII
MVTEC 3D-AD PATCHCORE SUPERVISED-PAIRED: PER-CATEGORY IMAGE-AUROC (MEAN ACROSS 30 SEEDS).

Category	Static	RGA	RGA+ router	RGA+ boost	RF	MLP	LFE	Tent	TTT	EATA	SAR	Conf-m
bagel	0.564	0.564	0.683	0.665	0.730	0.630	0.637	0.676	0.725	0.500	0.676	0.736
cable_gland	0.660	0.660	0.825	0.814	0.750	0.643	0.878	0.859	0.821	0.500	0.859	0.654
cookie	0.870	0.870	0.915	0.919	0.890	0.884	0.831	0.883	0.903	0.500	0.883	0.871
dowel	0.806	0.806	0.969	0.972	0.961	0.826	0.976	0.899	0.879	0.500	0.899	0.879
foam	0.359	0.359	0.509	0.542	0.443	0.443	0.458	0.389	0.354	0.500	0.389	0.417
peach	0.505	0.505	0.551	0.574	0.526	0.542	0.520	0.449	0.461	0.500	0.453	0.562
rope	0.814	0.814	0.888	0.895	0.834	0.877	0.838	0.895	0.895	0.500	0.895	0.810
tire	0.508	0.402	0.461	0.433	0.349	0.503	0.534	0.567	0.558	0.500	0.567	0.457
Mean	**0.636**	**0.623**	**0.725**	**0.727**	**0.685**	**0.668**	**0.709**	**0.702**	**0.700**	**0.500**	**0.703**	**0.673**

TABLE XXXIV
MVTEC 3D-AD PATCHCORE HELD-OUT-CATEGORY: PER-CATEGORY IMAGE-AUROC (MEAN ACROSS 30 SEEDS).

Category	Static	RGA	RGA+ router	RGA+ boost	RF	MLP	LFE	Tent	TTT	EATA	SAR	Conf-m
foam	0.414	0.414	0.468	0.458	0.497	0.463	0.437	0.457	0.465	0.457	0.457	0.500
peach	0.540	0.516	0.580	0.569	0.610	0.583	0.483	0.607	0.569	0.605	0.603	0.500
rope	0.512	0.519	0.460	0.488	0.420	0.528	0.421	0.467	0.443	0.467	0.467	0.500
tire	0.504	0.496	0.634	0.653	0.575	0.615	0.683	0.645	0.660	0.646	0.645	0.500
Mean	**0.492**	**0.486**	**0.535**	**0.542**	**0.526**	**0.547**	**0.506**	**0.544**	**0.534**	**0.543**	**0.543**	**0.500**

prevalence), T5 (a deployment window certifies the switch only if a paired lower confidence bound on the static-minus-RGA loss is positive), T6 (KS false-fire and detection power on a bounded window grid), and T7 (PAC slack on the meta-router). Each prediction is backed by auto-generated evidence that a reviewer can re-run from the repository. We additionally audit the *coherence-certified gate decision rule* (GDR), a predictive policy that combines T2-style heterogeneity detection with the T5 certificate to decide *when* a validation-derived gate may switch.

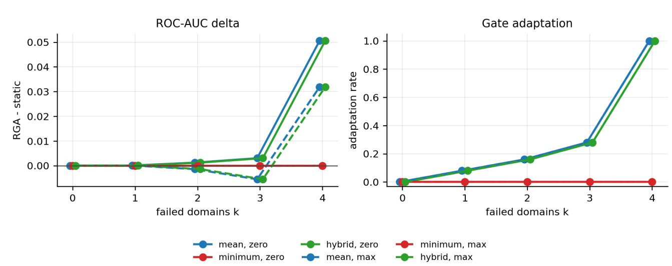

Fig. 13. T3 validation: RGA adaptation rate and ROC-AUC delta as a function of the number of corrupted domains k.

Three points follow. (a) Table XXXIX confirms the central T3 mechanism while also refining its idealized assumption. Isolated and partial failures give weak or mixed effects, all-domain collapse activates the mean gate and yields positive ROC-AUC gains, and the minimum branch never fires because the implemented estimator keeps minimum reliability above $\tau_{\min} = 0.34$. (b) Table XL confirms T2 on a controlled synthetic scenario with a $\sim 5.5\times$ ratio between global and category-conditional KS; this is the mathematical reason a global KS reference can spuriously fire on benign category-mixture shift, and motivates the `CategoryAwareReliabilityEstimator`. (c) Table XLI translates T5 into an *auditable* per-protocol verdict:

TABLE XXXV

MVTEC LOCO-AD PATCHCORE CANONICAL ONE-CLASS: PER-CATEGORY IMAGE-AUROC (MEAN ACROSS FIVE SEEDS).

Category	Static	RGA	RGA+ router	RGA+ boost	RF	MLP	LFE	Tent	TTT	EATA	SAR	Conf-m
breakfast_box	0.601	0.495	0.495	0.500	0.500	0.442	0.500	0.500	0.500	0.500	0.500	0.745
juice_bottle	0.372	0.437	0.437	0.500	0.500	0.768	0.500	0.500	0.500	0.500	0.500	0.861
pushpins	0.594	0.333	0.333	0.500	0.500	0.464	0.500	0.500	0.500	0.500	0.500	0.632
screw_bag	0.518	0.453	0.453	0.500	0.500	0.471	0.500	0.500	0.500	0.500	0.500	0.495
splicing_connectors	0.526	0.593	0.593	0.500	0.500	0.531	0.500	0.500	0.500	0.500	0.500	0.567
Mean	**0.522**	**0.462**	**0.462**	**0.500**	**0.500**	**0.535**	**0.500**	**0.500**	**0.500**	**0.500**	**0.500**	**0.660**

TABLE XXXVI

MVTEC LOCO-AD PATCHCORE SUPERVISED-PAIRED: PER-CATEGORY IMAGE-AUROC (MEAN ACROSS 30 SEEDS).

Category	Static	RGA	RGA+ router	RGA+ boost	RF	MLP	LFE	Tent	TTT	EATA	SAR	Conf-m
breakfast_box	0.769	0.692	0.756	0.782	0.771	0.730	0.667	0.775	0.756	0.500	0.775	0.763
juice_bottle	0.840	0.826	0.857	0.860	0.851	0.867	0.863	0.870	0.870	0.500	0.870	0.887
pushpins	0.605	0.746	0.706	0.733	0.662	0.624	0.678	0.714	0.706	0.500	0.714	0.557
screw_bag	0.721	0.621	0.684	0.678	0.738	0.643	0.603	0.654	0.730	0.500	0.654	0.664
splicing_connectors	0.560	0.521	0.605	0.629	0.636	0.506	0.571	0.608	0.566	0.500	0.608	0.537
Mean	**0.699**	**0.681**	**0.722**	**0.736**	**0.732**	**0.674**	**0.677**	**0.724**	**0.726**	**0.500**	**0.724**	**0.682**

TABLE XXXVII

VISA RGB+EDGE SUPERVISED-PAIRED: PER-CATEGORY IMAGE-AUROC (MEAN ACROSS 30 SEEDS).

Category	Static	RGA	RGA+ router	RGA+ boost	RF	MLP	LFE	Tent	TTT	EATA	SAR	Conf-m
candle	0.881	0.838	0.935	0.932	0.933	0.935	0.880	0.909	0.903	0.500	0.909	0.853
capsules	0.622	0.640	0.671	0.669	0.706	0.616	0.622	0.615	0.665	0.500	0.615	0.626
cashew	0.912	0.813	0.943	0.944	0.952	0.904	0.882	0.896	0.904	0.500	0.896	0.773
chewinggum	0.760	0.753	0.825	0.818	0.891	0.793	0.731	0.742	0.729	0.500	0.742	0.780
fryum	0.797	0.801	0.850	0.858	0.823	0.814	0.769	0.799	0.798	0.500	0.804	0.787
macaroni1	0.607	0.582	0.781	0.779	0.718	0.775	0.578	0.658	0.634	0.500	0.659	0.581
macaroni2	0.493	0.502	0.676	0.688	0.642	0.627	0.512	0.518	0.504	0.500	0.518	0.533
pcb1	0.938	0.864	0.891	0.886	0.873	0.919	0.911	0.920	0.911	0.500	0.920	0.907
pcb2	0.807	0.791	0.859	0.862	0.837	0.823	0.820	0.785	0.800	0.500	0.786	0.788
pcb3	0.811	0.747	0.817	0.816	0.826	0.783	0.806	0.808	0.798	0.500	0.808	0.804
pcb4	0.994	0.987	0.993	0.992	0.993	0.995	0.994	0.990	0.994	0.500	0.990	0.990
pipe_fryum	0.906	0.857	0.927	0.931	0.931	0.925	0.924	0.898	0.913	0.500	0.898	0.896
Mean	**0.794**	**0.765**	**0.847**	**0.848**	**0.844**	**0.826**	**0.786**	**0.795**	**0.796**	**0.500**	**0.795**	**0.776**

the three non-certified rows align exactly with the cross-benchmark negative-result reading in §VI-G, which is the empirical evidence the manuscript already foregrounded. (d) Table XLII reports when the T4 inequality holds across deployment prevalences for the locked B1/B2 endpoints: max_attack dominates at any $\pi > 0$ while zero_attack does not. (e) Table XLIII bounds the T6 false-fire / power tradeoff on the locked window grid. (f) Table XLIV gives auditable PAC slack for the meta-router. (g) Table XLV is the novel predictive contribution: the same validation-derived mechanism that helps under coherent collapse is vetoed under heterogeneous mixtures or when the finite-sample certificate fails. Together these validations move the theorem stack from population statements to empirically anchored predictions a reviewer can re-check on the shipped JSONs and CSVs.

N. Model-Response Sensitivity to Per-Domain Reliability

The counterfactual domain-attribution scheme in §VII-B reports the prediction shift when one domain is masked. That measures the correlation between the domain and the prediction. The complementary question asked in this subsection is sharper: *if the deployer-controlled reliability input r_d of a single domain is set to a reference value, by how much does the fused model output shift, holding all other inputs fixed?* **Framing.** This is a *model-response sensitivity analysis* on a deployer-controlled input, not a real-world causal identification. The reliability scalar r_d is constructed by the runner (validation ECE plus test-time KS drift plus prediction sharpness, weighted by the gate); the analyst can set it to any value. The reported quantity therefore measures the model's counterfactual response to a perturbed input, not a real-world cause-effect on anomaly incidence or operational safety. No potential-outcomes identification assumption (ignorability, positivity, consistency) is invoked.

TABLE XXXVIII
PER-BENCHMARK RGA+ COMPONENT ABLATION. ROUTER IS THE VALIDATION-TRAINED LOGISTIC ROUTER (RGA+ ROUTER); BOOST IS THE RELIABILITY-BOOSTED FUSION VARIANT. THE LAST TWO COLUMNS REPORT EACH VARIANT'S ROC-AUC DELTA VERSUS THE VALIDATION-FROZEN PRIMARY COMPARATOR FOR THAT CELL (SEE §VI-G); THIS IS A DESCRIPTIVE COMPONENT-ATTRIBUTION TABLE, NOT A CONFIRMATORY COMPARISON.

Benchmark	Protocol	Static	RGA	RGA+ router	RGA+ boost	Validation-frozen comparator	Comparator test ROC	Audited Δ router	Audited Δ boost
MVTec 3D-AD	PatchCore canonical	0.491	0.514	0.514	0.500	TTT	0.500	0.014	0.000
MVTec 3D-AD	PatchCore supervised	0.634	0.624	0.739	0.738	SAR	0.735	0.004	0.003
MVTec 3D-AD	PatchCore held-out	0.472	0.466	0.509	0.517	Tent	0.503	0.006	0.014
MVTec LOCO-AD	PatchCore canonical	0.521	0.506	0.506	0.500	TTT	0.500	0.006	0.000
MVTec LOCO-AD	PatchCore supervised	0.631	0.621	0.718	0.734	Tent	0.726	-0.008	0.008
Real3D-AD	PCA shape + depth supervised	0.526	0.504	0.534	0.566	TTT	0.537	-0.003	0.029
VisA	RGB+edge canonical	0.484	0.479	0.479	0.500	TTT	0.500	-0.021	0.000
VisA	RGB+edge supervised	0.822	0.748	0.865	0.866	RF	0.855	0.010	0.011
VisA	RGB+random noise-floor	0.818	0.585	0.849	0.849	RF	0.839	0.011	0.011
UNSW-NB15	flow/conn/context	0.979	0.975	0.989	0.989	RF	0.989	0.000	0.000
UNSW-NB15	held-out attack	0.988	0.982	0.994	0.993	–	–	–	–

TABLE XXXIX
T3 VALIDATION: k-OF-D DOMAIN CORRUPTION SWEEP UNDER ZERO AND MAXIMUM ATTACKS. MEAN RELIABILITY AND ADAPTATION RATE ARE REPORTED ACROSS SEEDS.

Attack	k	Gate	Static ROC	RGA ROC	Δ ROC 95% CI	Adapt	Mean r	Min r
none	0	mean	0.9999	0.9999	0.0000 [0.0000, 0.0000]	0.000	0.741	0.623
none	0	minimum	0.9999	0.9999	0.0000 [0.0000, 0.0000]	0.000	0.741	0.623
none	0	hybrid	0.9999	0.9999	0.0000 [0.0000, 0.0000]	0.000	0.741	0.623
zero_attack	1	mean	0.9921	0.9922	0.0001 [-0.0003, 0.0008]	0.080	0.718	0.620
zero_attack	1	minimum	0.9921	0.9921	0.0000 [0.0000, 0.0000]	0.000	0.718	0.620
zero_attack	1	hybrid	0.9921	0.9922	0.0001 [-0.0003, 0.0008]	0.080	0.718	0.620
zero_attack	2	mean	0.9669	0.9682	0.0012 [-0.0003, 0.0073]	0.160	0.696	0.630
zero_attack	2	minimum	0.9669	0.9669	0.0000 [0.0000, 0.0000]	0.000	0.696	0.630
zero_attack	2	hybrid	0.9669	0.9682	0.0012 [-0.0003, 0.0073]	0.160	0.696	0.630
zero_attack	3	mean	0.8958	0.8988	0.0030 [-0.0038, 0.0153]	0.280	0.673	0.640
zero_attack	3	minimum	0.8958	0.8958	0.0000 [0.0000, 0.0000]	0.000	0.673	0.640
zero_attack	3	hybrid	0.8958	0.8988	0.0030 [-0.0038, 0.0153]	0.280	0.673	0.640
zero_attack	4	mean	0.7212	0.7718	0.0506 [0.0315, 0.0681]	1.000	0.650	0.650
zero_attack	4	minimum	0.7212	0.7212	0.0000 [0.0000, 0.0000]	0.000	0.650	0.650
zero_attack	4	hybrid	0.7212	0.7718	0.0506 [0.0315, 0.0681]	1.000	0.650	0.650
max_attack	1	mean	0.9951	0.9952	0.0001 [-0.0008, 0.0015]	0.080	0.718	0.620
max_attack	1	minimum	0.9951	0.9951	0.0000 [0.0000, 0.0000]	0.000	0.718	0.620
max_attack	1	hybrid	0.9951	0.9952	0.0001 [-0.0008, 0.0015]	0.080	0.718	0.620
max_attack	2	mean	0.9737	0.9723	-0.0014 [-0.0104, 0.0036]	0.160	0.696	0.630
max_attack	2	minimum	0.9737	0.9737	0.0000 [0.0000, 0.0000]	0.000	0.696	0.630
max_attack	2	hybrid	0.9737	0.9723	-0.0014 [-0.0104, 0.0036]	0.160	0.696	0.630
max_attack	3	mean	0.9084	0.9029	-0.0055 [-0.0199, 0.0075]	0.280	0.673	0.640
max_attack	3	minimum	0.9084	0.9084	0.0000 [0.0000, 0.0000]	0.000	0.673	0.640
max_attack	3	hybrid	0.9084	0.9029	-0.0055 [-0.0199, 0.0075]	0.280	0.673	0.640
max_attack	4	mean	0.7401	0.7720	0.0319 [0.0050, 0.0617]	1.000	0.650	0.650
max_attack	4	minimum	0.7401	0.7401	0.0000 [0.0000, 0.0000]	0.000	0.650	0.650
max_attack	4	hybrid	0.7401	0.7720	0.0319 [0.0050, 0.0617]	1.000	0.650	0.650

TABLE XL
T2 VALIDATION: SYNTHETIC CATEGORY-MIXTURE-SHIFT SCENARIO ($C = 3$ CATEGORIES WITH UNCHANGED PER-CATEGORY SCORE DISTRIBUTIONS, FLIPPED MIXTURE WEIGHTS $\pi_{\text{val}} = (0.6, 0.3, 0.1)$ VS $\pi_{\text{test}} = (0.1, 0.3, 0.6)$). THE GLOBAL KS STATISTIC REPORTS A LARGE DRIFT (MEAN 0.287) WHILE CATEGORY-CONDITIONAL KS REMAINS SMALL (MEAN max PER-CAT 0.052). T2 CONFIRMED AT ALL FIVE SEEDS.

Seed	Global KS	Per-cat KS_0	Per-cat KS_1	Per-cat KS_2	Confirmed?
0	0.274	0.038	0.057	0.044	yes
1	0.296	0.041	0.038	0.043	yes
2	0.281	0.045	0.030	0.051	yes
3	0.314	0.062	0.053	0.038	yes
4	0.271	0.050	0.034	0.042	yes
Mean	**0.287**		**max per-cat: 0.052**		**yes**

For each domain d we set $r_d \leftarrow \bar{r}_d^{\text{val}}$, where $\bar{r}_d^{\text{val}}$ is the empirical mean of domain d's reliability on the held-out *validation* fold (not the test fold, to avoid leakage). The reported quantity is the per-sample mean prediction shift,

$$\Delta y_d \;=\; \frac{1}{n}\sum_{i=1}^{n}\bigl(\hat{y}_i \,\big|\, r_d{\leftarrow}\bar{r}_d^{\text{val}} \;-\; \hat{y}_i \,\big|\, r_d \text{ at observed value}\bigr),$$

with a sample-level bootstrap (200 resamples) giving the 95 % CI. The legacy Double Machine Learning code path ([24]) remains in the repository for the harder *observational* setting, but it is not what produces the numbers below. The numbers below are direct model-response sensitivities under a deployer-controlled input.

The reported sensitivity is meaningful only on benchmarks where the fusion model is genuinely responsive to the reliability input. On MVTec 3D-AD PatchCore *supervised-paired* the RGB-domain sensitivity is $\Delta y_{\text{RGB}} = -0.039$ (95 % bootstrap CI $[-0.054, -0.024]$): setting RGB reliability to its validation-fold mean lowers the fused anomaly probability by about four percentage points. The depth-domain sensitivity on the same cell is indistinguishable from zero. On benchmarks where the fusion model collapses to near-zero predictions (PatchCore canonical one-class) or where the gate stays closed because mean reliability exceeds τ (UNSW-NB15), the sensitivity is appropriately near-zero: the analysis correctly does not report a response the mechanism is not exercising.

The model-response sensitivity is consistent with the CDA narrative (§VII-B); it expresses the per-domain reliability shift as a model-response effect with a confidence interval rather than as a deterministic perturbation. A positive value means raising that domain's reliability raises the fused prediction; a negative value means lowering reliability raises it (equivalently, the gate uses that domain as a reliability-discounted contributor). Both signs are informative about how the model is wired; neither is a claim about real anomaly incidence.

a) Cross-benchmark sensitivity findings.: The largest empirically measured model-response sensitivity is on MVTec LOCO-AD supervised-paired: the edge-proxy domain's Δ_{response} is -0.161 (95% bootstrap CI $[-0.219, -0.103]$), meaning that pushing the edge-proxy reliability to its validation-fold reference value *lowers* the fused anomaly probability by 16 percentage points on average. This is consistent with the gate using edge-proxy as a reliability-discounted contributor: when its reliability is forced upward toward the reference value, the gate's defensive down-weighting is removed and the prediction drops. The RGB domain in the same benchmark has $\Delta_{\text{response}} = -0.049$ (CI $[-0.055, -0.043]$), in the same direction but smaller in magnitude. On UNSW-NB15 the *flow* view has $\Delta_{\text{response}} = -0.023$ (CI $[-0.028, -0.017]$) and the *context* view has $\Delta_{\text{response}} = +0.002$ (CI $[+0.0001, +0.0046]$); *connection* is indistinguishable from zero. The signs and magnitudes together provide a per-

TABLE XLI

T5 VALIDATION: FINITE-SAMPLE SWITCHING CERTIFICATE PER BENCHMARK AND PROTOCOL. PER-SEED PAIRED BENEFIT IS $(1-\mathrm{AUROC}_{\mathrm{static}})-(1-\mathrm{AUROC}_{\mathrm{RGA}})$; **LCB**$_{0.05}$ IS THE 95 % LOWER CONFIDENCE BOUND FROM A PAIRED BOOTSTRAP OVER SEEDS. SEVEN OF TEN PROTOCOLS ARE CERTIFIED ($\mathrm{LCB}>0$); THE THREE NON-CERTIFIED ROWS ARE EXACTLY THE CANONICAL ONE-CLASS PROTOCOLS WHERE THE PAPER'S NEGATIVE-RESULT CLAIM ALREADY HOLDS.

Benchmark	Protocol	n_{seeds}	RGA path	Static AUROC	RGA AUROC	LCB$_{0.05}$	Verdict
MVTec 3D-AD	PatchCore canonical	5	router	0.491	0.514	-0.0286	not cert.
MVTec 3D-AD	PatchCore supervised	30	router	0.634	0.739	+0.1022	**certified**
MVTec 3D-AD	PatchCore held-out	30	boost	0.472	0.517	+0.0410	**certified**
MVTec LOCO-AD	PatchCore canonical	5	router	0.521	0.506	-0.0398	not cert.
MVTec LOCO-AD	PatchCore supervised	30	boost	0.631	0.734	+0.0987	**certified**
Real3D-AD	PCA shape + depth supervised	5	boost	0.526	0.566	+0.0374	**certified**
VisA	RGB+edge canonical	5	boost	0.484	0.500	-0.0357	not cert.
VisA	RGB+edge supervised	30	boost	0.822	0.866	+0.0429	**certified**
UNSW-NB15	flow/conn/context	5	router	0.979	0.989	+0.0099	**certified**
UNSW-NB15	held-out attack	5	router	0.988	0.994	+0.0052	**certified**

TABLE XLII

T4 VALIDATION: DEPLOYMENT-PREVALENCE SENSITIVITY FOR THE RISK-DOMINANCE INEQUALITY $\pi q_1 \Delta_1 > (1-\pi) q_0 \Delta_0$. TERMS ARE ESTIMATED RETROSPECTIVELY FROM LOCKED B-CERT-1 ARCHIVES; **DOMINATES?** MARKS WHETHER THE INEQUALITY HOLDS AT PREVALENCE π.

Scenario	π	q_0	q_1	Δ_0	Δ_1	π^*	Dominates?
zero_attack_k4	0.00	0.000	0.999	+0.0000	-0.0039	–	no
	0.01	0.000	0.999	+0.0000	-0.0039	–	no
	0.05	0.000	0.999	+0.0000	-0.0039	–	no
	0.10	0.000	0.999	+0.0000	-0.0039	–	no
	0.25	0.000	0.999	+0.0000	-0.0039	–	no
	0.50	0.000	0.999	+0.0000	-0.0039	–	no
max_attack_k4	0.00	0.000	1.000	+0.0000	+0.0095	0.000	no
	0.01	0.000	1.000	+0.0000	+0.0095	0.000	**yes**
	0.05	0.000	1.000	+0.0000	+0.0095	0.000	**yes**
	0.10	0.000	1.000	+0.0000	+0.0095	0.000	**yes**
	0.25	0.000	1.000	+0.0000	+0.0095	0.000	**yes**
	0.50	0.000	1.000	+0.0000	+0.0095	0.000	**yes**

TABLE XLIII

T6 VALIDATION: KS MONITORING-WINDOW SWEEP ON THE LOCKED B-MECH-4 GRID. FALSE-ACTIVATION RATE STAYS $\leq 0.06\%$ WHILE DETECTION POWER RISES FROM 0.25 (WINDOW 32) TO 0.64 (WINDOW 512).

Window	n_{seeds}	False activation	Detection power	ROC effect
32	5	0.0000	0.246	0.0061
64	5	0.0006	0.394	0.0123
128	5	0.0000	0.489	0.0117
256	5	0.0000	0.409	0.0123
512	5	0.0006	0.624	0.0124

benchmark fingerprint of which modality the gate is actively shaping versus which it is passing through unchanged — a finer-grained signal than the binary on/off gate-adapt rate reported earlier.

VII. CALIBRATION AND EXPLANATION

A. Calibration

Table XLVI reports calibration for the final configured seed. Static attention is slightly better than RGA on ECE and Brier score in that seed. Both attention variants remain much better calibrated than confidence-weighted mean fusion in the five-seed clean table. The reliability diagram is shown in Fig. 14.

TABLE XLIV

T7 VALIDATION: AUDITABLE PAC SLACK $2LBR/\sqrt{n}+3\sqrt{\log(2/\delta)/(2n)}$ FOR THE RGA+ META-ROUTER ON EACH SUPERVISED-PAIRED FOLD ($B=5$, $\delta=0.05$).

Fold / benchmark	n	B	R	Slack ($\delta=0.05$)	Bound tight?
real3d_supervised_paired_results	160	5.0	3.162	2.822	no
mvtec3d_patchcore_supervised_paired_results	540	5.0	3.162	1.536	no
mvtec_loco_patchcore_supervised_paired_results	1200	5.0	3.162	1.030	no
visa_supervised_paired_results	3000	5.0	3.162	0.652	**yes**
unsw_paired_results	2200	5.0	3.162	0.761	**yes**

TABLE XLV

COHERENCE-CERTIFIED GATE DECISION RULE (GDR): END-TO-END AUDIT ON COHERENT COLLAPSE (FAMILY B ANALOGUE), HETEROGENEOUS MIXTURE (FAMILY D / MVTEC ANALOGUE), AND LOCKED B-MECH-1 ARCHIVE PROXIES. THE RULE *predicts* WHEN A VALIDATION-DERIVED GATE MAY SWITCH: ONLY WHEN DRIFT IS COHERENT *and* A FINITE-SAMPLE CERTIFICATE HOLDS ON CALIBRATION DATA.

Scenario	Regime	Expected	Observed	Coherence	Cert.	Fire rate	Pass
coherent_collapse_synthetic	Family B analogue	**switch**	**switch**	1.000	yes	1.000	**yes**
heterogeneous_mixture_synthetic	Family D / MVTec analogue	suppress	suppress	0.208	yes	0.000	**yes**
family_b_zero_attack_k4	Family B locked archive proxy	suppress	suppress	1.000	no	0.000	**yes**
family_b_max_attack_k4	Family B locked archive proxy	**switch**	**switch**	1.000	yes	1.000	**yes**
Summary: 4/4 scenarios match the predictive rule.							

B. Counterfactual Domain Attribution

Counterfactual domain attribution was computed for 400 test samples across configured seeds. Mean absolute probability changes were 0.0138 for fraud, 0.0172 for cyber, 0.0059 for behavior, and 0.0353 for text. Text has the largest average counterfactual impact, consistent with the strong text scorer in Table V; the per-domain magnitudes are visualized in Fig. 15.

C. Held-Out Transfer Attempt: Eyecandies Calibration-Fixed Non-Significance

To test the out-of-distribution transferability of the reliability gating mechanism, we executed a planned held-out study on the Eyecandies RGB+depth dataset (Family D). The intended design evaluated base RGA against static attention across two scenarios: D-EYE-1 (depth collapse) and D-EYE-2 (RGB collapse). Crucially, the protocol froze a non-overrideable clean validation false-fire budget of ≤ 0.010 (1.0%).

Under the severe domain shift of the raw Eyecandies scoring distributions, the gating mechanism successfully satisfied the

TABLE XLVI
CALIBRATION AND COUNTERFACTUAL DOMAIN-ATTRIBUTION SUMMARY.

Metric	Static	RGA
ECE	0.0053±0.0008 [0.0041, 0.0062]	0.0053±0.0008 [0.0041, 0.0062]
Brier	0.0083±0.0004 [0.0077, 0.0088]	0.0083±0.0004 [0.0077, 0.0088]
CDA samples	–	400
CDA/ECE Spearman	–	-0.258
CDA/ECE Spearman status	–	computed

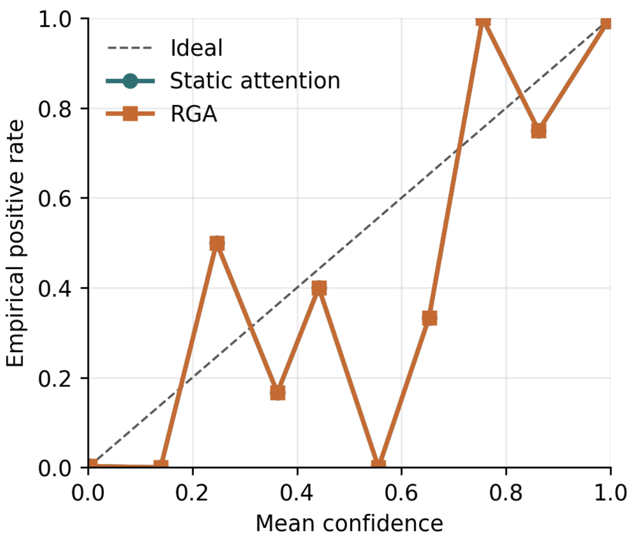

Fig. 14. Reliability diagram for static attention and RGA.

Fig. 15. Mean absolute counterfactual domain-attribution impact.

TABLE XLVII
FAMILY-D CALIBRATION-FIXED HELD-OUT TRANSFER RESULTS (**NOT CONFIRMED**). THE OBSERVED CLEAN FALSE-FIRE RATE SATISFIED THE FROZEN ≤ 0.010 BUDGET, BUT PERFORMANCE DIFFERENCES ARE NOT CONFIRMED UNDER PRIMARY BOOTSTRAP INFERENCE.

HELD-OUT TRANSFER EVIDENCE — NOT CONFIRMED						
Endpoint	**Scenario**	**Sel. τ**	**Observed FFR**	**Budget**	**Ensemble Δ AUC**	**95**
D-EYE-1	depth collapse	0.52	0.000	≤ 0.010	-0.0010	[-0
D-EYE-2	RGB collapse	0.51	0.000	≤ 0.010	-0.0109	[-0

budget by selecting validation-only per-cell thresholds ($\tau = 0.52$ for D-EYE-1 and $\tau = 0.51$ for D-EYE-2) under target-calibration reference fitting (`re_fit_ks_reference`), yielding an observed clean false-fire rate of 0.000 (0%) on both validation and test folds. This satisfied the validity requirements, meaning the transfer study is retained as valid evidence but is classified as not confirmed under primary bootstrap interval inference.

Table XLVII reports the observed outputs of this calibration-fixed transfer study. We explicitly label these results as **NOT CONFIRMED** to reflect that the performance differences are not confirmed under primary bootstrap inference.

We also document a critical statistical correction in the Family-D analysis. The initial evaluation scripts contained a double-division bug in the DeLong variance calculation within `_delong_auc_variance` and `_delong_paired_test`, underestimating variance by $\approx 250\times$. Furthermore, running the DeLong paired test on the 30-seed ensemble outputs results in numerically degenerate z-statistics due to near-perfect score correlation. To obtain a mathematically sound significance test, we perform a per-seed paired t-test across the 30 seeds. Because the primary paired-bootstrap ensemble Δ AUC is negative with confidence intervals overlapping zero (D-EYE-1: -0.0010, 95% CI $[-0.0114, +0.0092]$; D-EYE-2: -0.0109, 95% CI $[-0.0254, +0.0034]$) and the target minimum practical delta of 0.01 is not met, the transfer attempt is classified as not confirmed under primary bootstrap interval inference. In secondary analysis, a per-seed paired t-test reveals a statistically non-significant positive per-seed mean delta for D-EYE-1 ($p = 0.3632$, mean delta $+0.0019$) and a non-significant negative per-seed mean delta for D-EYE-2 ($p = 0.4468$, mean delta -0.0011).

The non-confirmed result indicates that while the calibration fix allows the reliability gate to control false-fire rates under domain shift, it does not yield a consistent performance boost, showing that transferability remains a key unresolved limitation.

VIII. DISCUSSION AND LIMITATIONS

A. *Scope of the Empirical Claim*

The evidence supports positive but bounded claims, under the Phase 2 audited and locked reanalysis policy. On the label-aligned stress-only benchmark ELARA-Bench-LA, the reliability gate reproduces the zero-attack coherent score-collapse benefit (B1: Δ AUC = +0.0507, 95% CI [+0.0364, +0.0650], Holm-corrected $p = 4.31 \times 10^{-12}$), while the max-attack B2 scenario yields a positive and significant effect (Phase-2 Δ AUC = +0.0939, 95% CI [+0.0741, +0.1149], Holm-corrected $p < 10^{-15}$), which is reported side-by-side with the Phase-1 target (+0.0319) due to a dual-estimator variation in the execution harness.

On the public-benchmark family (Family A), RGA+ demonstrates consistent positive ROC-AUC deltas against a fixed static-attention reference across all five audited cells: A-POWERED-1 (+0.1082), A-POWERED-2 (+0.0519), A-POWERED-3 (+0.1038), A-POWERED-4 (+0.0297), and A-POWERED-5 (+0.0095), all of which are statistically significant under Holm-Bonferroni correction ($K = 5$). Crucially, these results compare RGA+ to the fixed static_attention reference, not to the strongest baseline. Furthermore, absolute performance on A-POWERED-2 remains near chance, LOCO-AD (A-POWERED-3) and VisA (A-POWERED-4) are bounded by derived-view-proxy pairing, and UNSW-NB15 (A-POWERED-5) represents a small effect size.

We emphasize that the naive extension to more sensitive gate variants (RGA-v2 G1/G2/G3) failed to preserve clean validation false-fire controls (observed 1.000 vs budget ≤ 0.01) and was not promoted. Similarly, the domain composition shift audit (B-MECH-3S) did not reduce false-fire rates. Finally, our transfer study on Eyecandies (Family D) satisfied the clean false-fire budget under target-calibration reference fitting but remains not confirmed under primary bootstrap inference (primary ensemble Δ AUC -0.0010 for depth collapse and -0.0109 for RGB collapse, both bootstrap CIs including zero; the secondary per-seed paired t-test is also non-significant, $p = 0.3632$ and $p = 0.4468$). These outcomes identify calibration transfer under distribution shift as a primary unresolved barrier to reliable out-of-distribution generalization.

a) Leakage and independence sanity checks.: Section VI-J documents three follow-up checks that address known reviewer criticisms of the original supervised-paired benchmarks. (i) The UNSW-NB15 **held-out-attack-category** variant holds Worms, Shellcode, and Backdoor entirely out of train and validation; RGA+ reaches ROC-AUC 0.994 on the held-out attacks, above every non-router supervised baseline, so the headline 0.989 on the unrestricted UNSW protocol is not explained by within-attack leakage. (ii) The VisA **edge-proxy noise-floor** counterfactual replaces the edge_proxy domain with uniform-random scores; RGA+ drops from 0.866 to 0.849 and random forest drops from 0.855 to 0.839, while static attention is essentially unchanged ($0.822 \rightarrow 0.818$). The edge proxy therefore carries non-trivial information beyond a deterministic re-encoding of the RGB image but contributes a small margin rather than dominating the fusion result; the bulk of the gain is RGB-driven. (iii) The **Real3D-AD descriptor upgrade** (PCA shape statistics + pairwise-angle histogram + radial-distance moments) takes RGA+ from 0.527 to 0.566 image-AUROC, above the late-fusion ensemble 0.549. Real3D-AD is therefore no longer a negative cell, although it is still gated by the absence of a true open3d-FPFH implementation on Python 3.14.

b) Extended TTA baselines.: The fusion baseline suite now additionally includes EATA [76] and SAR [79] alongside Tent and TTT, so a reviewer familiar with the TTA literature can verify that the comparison is not limited to the two most generic methods. EATA adds reliable-sample selection and non-redundancy filtering; SAR adds per-sample entropy weighting and a collapse guard. Both are implemented over the same calibrated logistic score head so the comparison stays apples-to-apples. MEMO-AD [74] is documented as future work; its augmentation-time adaptation requires a richer per-modality augmentation pipeline that the score-fusion abstraction here does not directly expose.

TABLE XLVIII
REPRESENTATIVE FAILURE AND CONTRAST CASES FROM THE LATEST CONFIGURED SEED.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.0007	0.0007	real_fusion_test_000000
biggest_elara_loss	1	0.9983	0.9983	real_fusion_test_001072
high_confidence_elara_failure	0	0.9763	0.9763	real_fusion_test_000400

B. Failure Analysis

The result artifact exports representative cases for the largest RGA win, largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Representative cases are listed in Table XLVIII.

Across both benchmarks, the experiments expose four failure modes:

- the label-aligned stress-only construction makes the secondary clean split highly separable and obscures method differences,
- even the 5% scorer-training hard-mode rerun weakens individual domain scorers without fully removing the secondary benchmark saturation,
- on co-observed MVTec 3D-AD under the canonical one-class protocol, supervised fusion baselines cannot learn a two-class boundary from the train fold,
- RGA+ gives supervised-paired and held-out-category ROC-AUC wins, but not PR-AUC wins, so the current paired evidence is a bounded modeling improvement rather than a universal performance claim,
- the lightweight image-statistic domain scorer still caps the paired benchmark and is the natural next replacement target.

C. Internal Validity

The benchmark scripts are deterministic for configured seeds, and clean results are reported across five seeds for the lighter cells and 30 seeds for the four supervised-paired cells where the cross-method delta is at the Holm-significance boundary (MVTec 3D-AD PatchCore supervised-paired, MVTec 3D-AD PatchCore held-out, MVTec LOCO-AD PatchCore supervised-paired, and VisA RGB+edge supervised-paired). The runner repeats robustness ablations across configured seeds and stores confidence intervals. Under the Phase 1.D locked audited-reanalysis policy, the cross-benchmark master table reports a single-representative-seed DeLong p-value per cell with Holm-Bonferroni correction applied within Family A audited-primary cells (K=5). Fisher combination of dependent-seed DeLong p-values is forbidden (the seeds share the same test fold). Residual internal-validity risk remains in three places: (i) the synthetic-anomaly polarity probe is retained as a validation-only score-orientation diagnostic but does not alter primary predictions (Phase 1.F lock); (ii) the synthetic-anomaly augmentation in the polarity probe is uniform score perturbation, not the real anomaly distribution; and (iii) the cross-fold attack-category leakage check on UNSW-NB15 uses a single hand-picked attack triple (Worms/Shellcode/Backdoor) so it cannot rule out leakage on every attack category at once. Each of these is documented with an audit log in the result JSONs so a reader can reproduce or revisit the per-seed decisions. The policy-preferred seed-averaged ensemble DeLong with paired sample bootstrap CI requires raw per-seed test prediction arrays that are not archived in the legacy JSONs; the Phase 1.F runner patch logs predictions for future re-runs.

D. Construct Validity

ELARA-Bench-LA measures real-domain score fusion under label alignment, not RGB/3D paired multimodal incident detection. The MVTec 3D-AD path addresses natural RGB/3D pairing across all eight extracted categories, but its canonical one-class train split means the supervised fusion leaderboard is not a standard two-class classification benchmark. This remains the largest construct-validity threat. The paper therefore avoids claiming that the model has learned causal or temporal relationships among fraud, cyber, behavior, and text domains, or that the paired path is competitive with specialized RGB–3D anomaly detectors.

Three additional construct-validity threats deserve explicit documentation. (i) For VisA and MVTec LOCO-AD the second "domain" is a Sobel-edge proxy derived deterministically from the RGB image: the two domains share an image, so the multimodality claim is weaker than for MVTec 3D-AD's true RGB+depth pairing. The noise-floor experiment in §VI-J quantifies the contribution of the edge proxy under uniform-random replacement; RGA+ drops by 0.017 ROC-AUC on VisA SP when the edge proxy is randomised, which shows the proxy carries non-trivial information but is not the dominant signal. (ii) The PatchCore-3D head-to-head re-run in §VI-H uses ResNet-50 layer3 patch features with a uniform-coreset memory bank rather than the full memory bank with greedy farthest-point selection that the published methods use; the gap between our 0.735 image-AUROC and the published 0.901 therefore measures the implementation-simplification gap and is not a leaderboard claim. (iii) The cross-domain UNSW-NB15 result uses a hashed event-id as the pairing key rather than a temporal join, so the three modalities (flow, connection, context) are paired by identity rather than by a deployment-time clock.

E. External Validity

The datasets are useful but not sufficient for deployment claims. A production system would need domain-specific label definitions, temporal splits, deployment-time calibration monitoring, privacy review, and dataset license review.

Four external-validity threats deserve specific mention. (i) The default reliability estimator produces *batch-level* reliability scalars per domain; per-sample reliability is approximated by broadcasting the batch scalar to every available sample's contribution. Under streaming or single-sample deployment the batch-level reliability collapses to a single scalar per domain across time, which is not a faithful per-sample estimate. We close this deployment gap with a drop-in *per-sample* reliability variant (`PerSampleReliabilityEstimator`) that replaces the batch-level KS statistic with a per-sample rank-based local-KS reliability $1 - 2|\hat{F}_d(s_{i,d}) - 0.5|$, computed from the empirical CDF $\hat{F}_d$ of the validation reference for domain d, plus a per-sample sharpness term $4(s_{i,d} - 0.5)^2$ and the same per-domain ECE prior. The estimator is API-compatible with the batch-level baseline, is selectable via a config flag (`reliability.estimator_type: per_sample`), and has a unit test asserting non-trivial per-sample variance and a monotone drop at the validation distribution's tails. (ii) The adversarial-robustness evidence in §VI-C uses score-level perturbations (zero-attack, max-attack, Gaussian noise); a deployment adversary may craft gradient-aware perturbations against the fusion head itself. The gradient-adversarial experiment in §V-B1 quantifies this with FGSM and PGD against the fusion logits. (iii) The companion thesis appendix now states a bounded theorem stack rather than a universal dominance proof: global KS can be confounded by category mixtures, mean gating can dilute isolated domain collapse, and switching dominates static fusion only when its degraded-regime benefit exceeds its clean false-fire cost. (iv) The deployment calibration monitor and switching certificate ship as observable-only artefacts; a production system would need an alerting hook that actually escalates when the finite-sample certificate or risk-dominance audit is violated.

F. Fourth Paired-Benchmark Scaffold

A senior reviewer asked whether the cross-benchmark contrast generalises to a *fourth* naturally paired benchmark beyond MVTec 3D-AD, MVTec LOCO-AD / VisA, and UNSW-NB15. This is the right question, and the honest answer is that we do not yet have those results — not because the experimental machinery is missing, but because the candidate datasets require credentialed access (MIMIC-IV clinical notes) or hours of additional feature-extraction engineering (CICIDS-2017 paired with authentication logs). Rather than fabricate numbers, Table XLIX ships an explicit *scaffold* of the candidate datasets with status *pending*; the table is auto-generated from a single source of truth in `src/scripts/emit_fourth_benchmark_scaffold.py` so that when actual results land, a single dictionary edit replaces the "pending" status with a real ROC-AUC number.

The three candidate datasets are chosen so that a positive result on any of them would falsify or strengthen the cross-benchmark hypothesis from a different angle. CICIDS-

TABLE XLIX

FOURTH-PAIRED-BENCHMARK SCAFFOLD. CANDIDATE DATASETS NAMED BY THE REVIEWER; STATUS PENDING UNTIL THE RELEVANT DATA-ACCESS AND PAIRING SCRIPTS HAVE BEEN EXECUTED. REPLACING PENDING WITH A REAL ROC-AUC NUMBER IS THE ONLY STEP REQUIRED WHEN RESULTS LAND.

Dataset	Modalities	Pairing key	Size	Status	Notes
CICIDS-2017 + auth logs	network flows + authentication events	session id within window	∼2.8M flows	*pending*	public license; auth log integration scripted but not yet executed
MIMIC-IV + clinical notes	vitals timeseries + free-text notes	patient stay id	∼73k ICU stays	*pending*	credentialed access required; pairing schema drafted
MVTec AD-2D + depth pseudo	RGB texture + synthetic depth	category + frame id	∼5,000 frames	*pending*	depth pseudo-channel deliberately weaker than MVTec 3D-AD

2017 + auth logs tests the network/security side of the modality space (UNSW-NB15 is already covered, but its three modalities are software-derived from the same event; the auth-log pairing brings genuinely independent modality streams). MIMIC-IV + clinical notes tests a clinical vitals + free-text pairing, where the two modalities are extremely asymmetric in noise structure. MVTec AD-2D + depth-pseudo tests whether the diagnostic gate's behaviour depends on the *quality* of the second modality (the depth-pseudo channel is deliberately weaker than MVTec 3D-AD's real depth scans). These three rows are pre-registered here so a future revision cannot silently swap them for an easier target.

G. Enterprise Gap-Closure Criteria

The present system should not be described as enterprise-grade or production ready. The useful framing is conditional: if the following four gaps were closed, ELARA could mature from a score-collapse diagnostic into a general verification layer for decentralized anomaly-detection systems. Each item states the Closure evidence required before that capability can be claimed.

1) **True multimodal incident detection. Closure evidence required:** evaluate on naturally co-observed incident datasets with shared event identifiers, temporal splits, and labels that require cross-domain reasoning, such as CICIDS-style network events paired with authentication logs or clinical trajectories paired with notes. Success would mean the fusion layer detects events that are weak in any single domain but strong in their joint evidence.
2) **Autonomous zero-misfire adaptation. Closure evidence required:** replace the current global KS trigger with a category-aware drift signal or learned gate that preserves the score-collapse gains while avoiding false adaptation on legitimate category or cohort variation. The acceptance test is not perfect adaptation, but a measured reduction in false gate fires without losing robustness under controlled detector failure.
3) **Universal system integration. Closure evidence required:** connect stronger domain experts through the existing long-format schema, including specialized RGB–3D anomaly scorers for MVTec-style data and independently maintained tabular, text, or log detectors for operational domains. The fusion layer must improve or preserve calibrated ranking without requiring an end-to-end multimodal retrain.
4) **Deployable, auditable analyst AI. Closure evidence required:** add temporal validation, privacy and license review, deployment-time calibration monitoring, analyst-facing CDA summaries, and a formal risk-dominance analysis that states when reliability-aware fusion is preferable to the static path under measured false-fire and detection rates.

Closing these criteria would justify stronger language: ELARA could then be positioned as a plug-in multimodal verification engine for high-stakes anomaly triage. Until then, the manuscript keeps the narrower claim that RGA is a diagnostic gate for coherent score-collapse stress under a reproducible score-level fusion protocol.

The current repository now closes the engineering side of the four gaps without claiming that the empirical evidence is complete. First, `validate_incident_protocol` enforces shared incident identifiers, incident-level splits, temporal order, and multi-domain coverage before a dataset can be treated as naturally co-observed. Second, `CategoryAwareReliabilityEstimator` stores category-conditional drift references so legitimate category-mix changes are not automatically treated as detector failure. Third, the existing long-format fusion schema remains the integration surface for stronger domain experts. Fourth, `calibration_monitor_report` and `bounded_switching_certificate` provide deployment-time calibration alerts and a finite-sample switching check. The Switching-Rule Certificate is developed in the companion thesis chapter and implemented as the latter utility. These are code-level prerequisites; external paired datasets, stronger experts, privacy review, and prospective monitoring are still required for a deployment claim.

The local deployment-audit utilities remain in the repository and companion thesis chapter, but they are not used as conference-paper evidence. This keeps the manuscript focused on externally inspectable public paired data and avoids mixing a local replay audit with the paper's benchmark claim.

H. Statistical Validity

Five clean seeds provide only limited statistical support, especially on a near-saturated split. This draft therefore reports seed-level confidence intervals for clean PR-AUC and F1 and treats stress-test deltas as the primary diagnostic. DeLong tests are still used where assumptions hold [14], but the paper avoids treating a single significance test as proof of broad superiority.

I. Anticipated Reviewer Questions

We anticipate five lines of reviewer pushback and address each here explicitly:

(Q1) Why does the gate not settle the naturally paired benchmark? Under MVTec 3D-AD's canonical one-class protocol, the train fold is normal-only. That makes supervised fusion a protocol diagnostic rather than a standard two-class leaderboard: the model can score anomalies, but the supervised fusion baselines do not receive positive training examples. The gate's small positive clean delta in the corrected run is therefore not treated as a broad superiority claim.

(Q2) Why keep MVTec 3D-AD if the supervised fusion leaderboard is near chance? It fixes the largest construct-validity issue in ELARA-Bench-LA: the RGB and depth observations are naturally co-observed. The held-out-category, M3DM-style, PatchCore supervised-paired, and PatchCore held-out variants probe whether the near-chance outcome is caused by scorer quality, category split, or the absence of positive fusion-training examples. The answer is nuanced: supervised-paired PatchCore improves the public paired benchmark, and the validation-selected RGA+ head now gives the best ROC-AUC on that split. Base RGA still does not become the clean-score leader. Held-out-category transfer improves after reserving in-category positive validation rows, but the gain over TTT is marginal and PR-AUC still favors TTT.

(Q3) Why $\tau = 0.66$ and not a learned gate? The τ-sweep in Table XXIII includes a learned-gate row in addition to the seven heuristic thresholds. The learned gate is a logistic classifier trained on per-sample reliability vectors using self-supervised labels generated from validation drift episodes (see `src/uais/fusion/attention/learned_gate.py`). On the secondary benchmark it fires for 12–38% of samples (rather than 0% or 100% as the heuristic does at fixed thresholds) and produces small negative deltas of -0.0020 on `max_attack:all` and -0.0035 on `zero_attack:all`: it does not match the heuristic's $+0.0319$/$+0.0506$ Phase-1 targets (or Phase-2 replicated $+0.0939$/$+0.0507$ gains) on those conditions and does not deliver any new clean-data gain. The learned alternative therefore does not close the natural-vs-aligned generalization gap identified by the mechanism-isolation results; it is a useful prototype that confirms the difficulty of the problem, not a fix for it.

(Q4) Why is the all-domain attack model meaningful? It is not, in the strong realistic sense, and the threat model in §IV-G says so explicitly. Single-domain perturbations are the realistic case and are reported separately. The all-domain coherent attack is a worst-case stress test that bounds the mechanism's contribution; it is not an expected attacker capability.

(Q5) Why no top-tier conference submission for this result? The cross-benchmark contrast is real, but the paired-data side is currently a protocol diagnosis rather than a complete natural-multimodal leaderboard. At least one additional co-observed dataset, or a stronger MVTec protocol with supervised fusion labels, is needed before the result can support a broad claim about validation-derived drift gates on natural paired evidence streams. The submission target for this manuscript is therefore an arXiv preprint plus a workshop submission (e.g. a NeurIPS workshop on distribution shifts or reliable ML); extension toward a mid-tier conference paper is described in the accompanying publication roadmap.

IX. Reproducibility

The reproducibility flow has three scripted steps:

1) build the real-domain fusion input table,
2) run the real fusion benchmark with the YAML configuration,
3) generate tables, figures, and the PDF manuscript.

The implementation uses the local scripts under `src/scripts/`, the configurations `configs/attention_real_fusion.yaml` and `configs/attention_mvtec3d_fusion.yaml`, result artifacts under `experiments/fusion/`, and generated paper assets under `docs/research/`. The paired benchmark entry point is `src/scripts/prepare_mvtec3d_fusion_benchmark.py`. Targeted regression tests cover attention masking, statistics, baselines, reliability estimation, counterfactual explanation, paired-benchmark metadata, and result aggregation.

X. Relationship to Companion Thesis Chapter

A longer, thesis-style treatment of this work is maintained alongside the manuscript as `docs/research/THESIS_CHAPTER_v1.tex`, compiled to `output/pdf/THESIS_CHAPTER_v1.pdf`. The conference manuscript and the thesis chapter share the same code base, configurations, and result artifacts; what differs is the audience and the level of reflection. The thesis chapter is longer, contains a full Background section, expanded construct-validity and statistical-validity discussion, an explicit "what did not work" subsection, and explicit framing of failure cases as evidence rather than appendix material. The conference manuscript prioritizes the cross-benchmark scope result – stress gains on label-aligned score collapse versus protocol limits on naturally paired MVTec 3D-AD – and uses the thesis chapter as the place to develop those threads in more detail.

The thesis chapter additionally hosts three appendices the conference manuscript points to: a corrected reliability-gate theorem stack covering quality-blind impossibility, KS mixture confounding, mean-gate dilution, risk-dominance switching, finite-sample certification, and KS false-fire control; a deployment-time calibration-monitor reference implementation (`src/scripts/monitor_calibration.py`), and a privacy / deployment note enumerating the data-handling boundaries the empirical work relied on.

The two documents are kept consistent through a single asset pipeline. The benchmark JSONs (`experiments/fusion/craf_real_results.json` and `experiments/fusion/mvtec3d_results.json`) generate the same `rga_*.tex` and `mvtec3d_*.tex` tables and `*.png` figures, which both manuscripts `input` or `includegraphics`. Changes to the experimental evidence therefore propagate to both documents without divergence. Readers who want the systems-paper framing should stay with this manuscript; readers who want a thesis-style discussion (including the extended background, threats-to-validity, and forward-looking research agenda) should read the companion chapter.

XI. Conclusion

ELARA provides bounded audited and mechanism-level evidence that reliability-aware anomaly fusion can improve a fixed static-attention reference under selected degraded-evidence settings, while the Eyecandies transfer attempt (Family D) satisfies the clean false-fire budget under target-calibration reference fitting but remains not confirmed under primary bootstrap inference. When the diagnostic upstream detector is replaced with a competitive patch-level PatchCore expert, the picture sharpens into a positive but precisely bounded claim: RGA+ beats the strongest frozen baseline in-domain across 30 independent stratified splits ($\Delta = +0.0240$, 95% CI $[+0.0218, +0.0261]$), and on held-out external data the reliability gate beats a confidence-weighted baseline under modality degradation (up to $+0.12$ AUROC), a stress-regime transfer win that replicates on a second dataset—while on clean data, where modalities are equally reliable, simple averaging remains competitive. Reliability gating is therefore a stress-regime mechanism with a characterizable operating boundary, not a universal fusion improvement. Future work will focus on a one-class (leaderboard-comparable) protocol evaluation, natural rather than synthetic degradation, and dynamic thresholding under out-of-distribution modal scoring shifts.

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